Agentic AI and Workflow Automation

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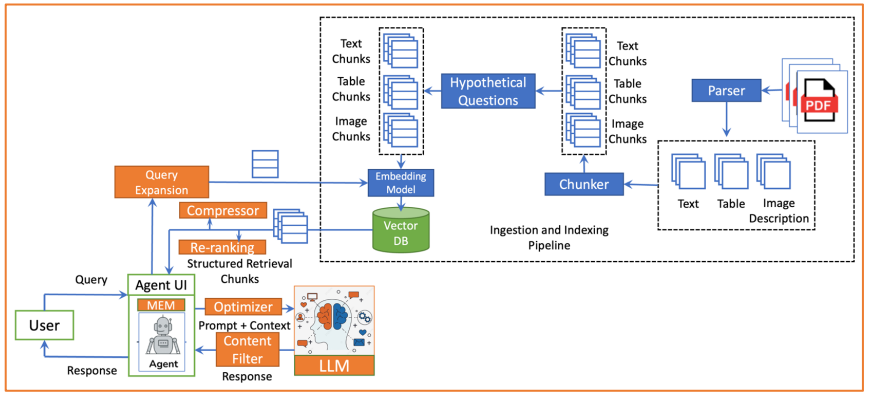
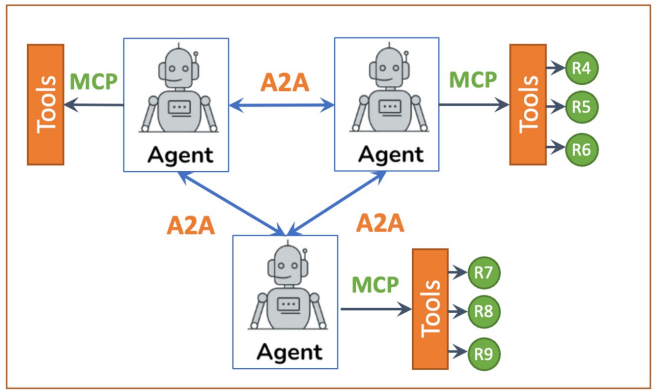
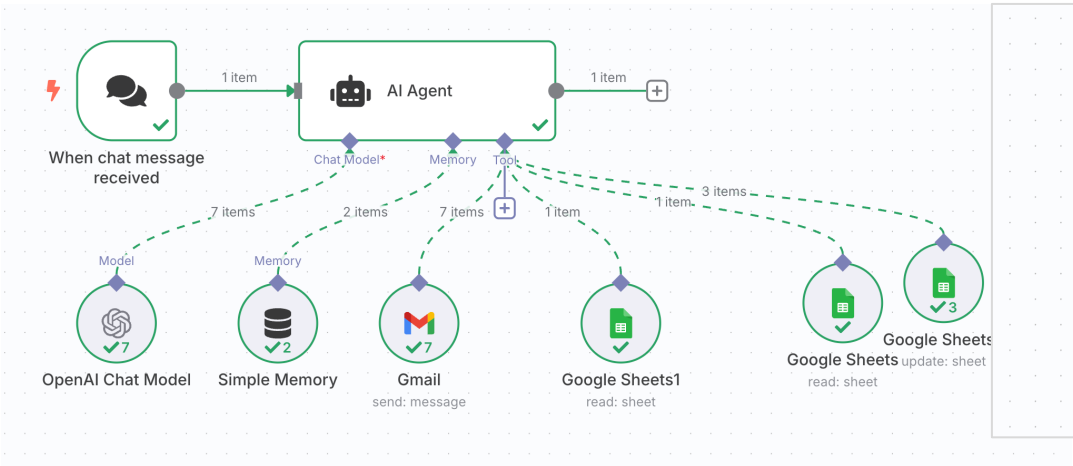
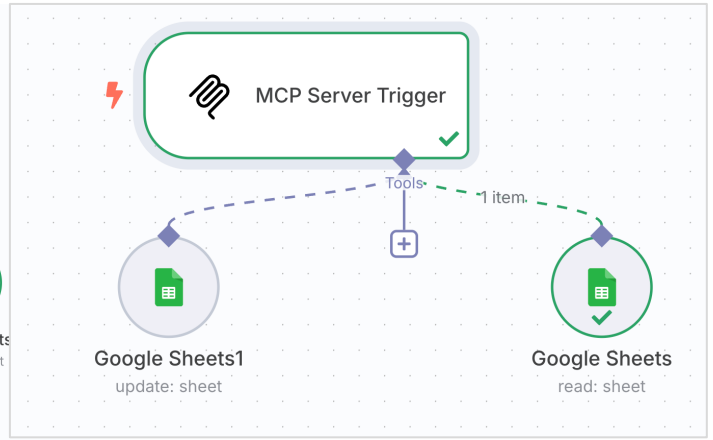








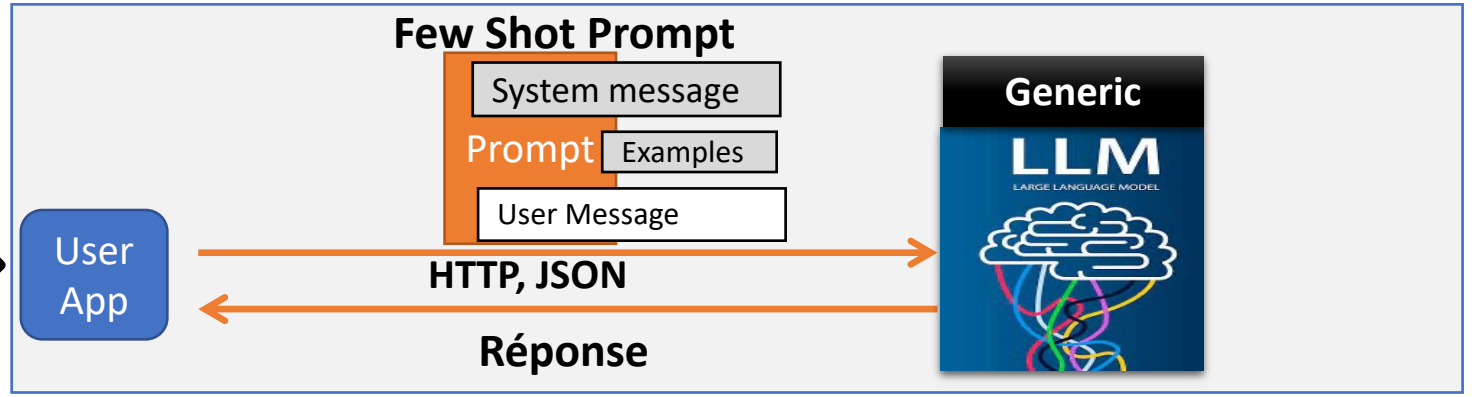
Leverage Generative AI

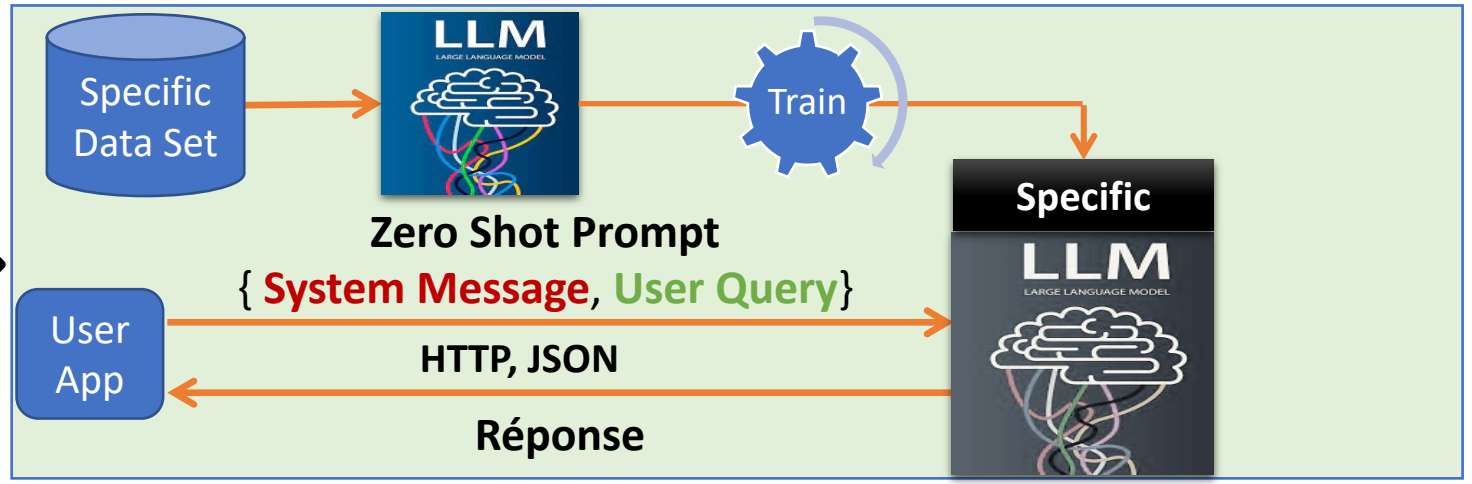
How to leverage Generative AI

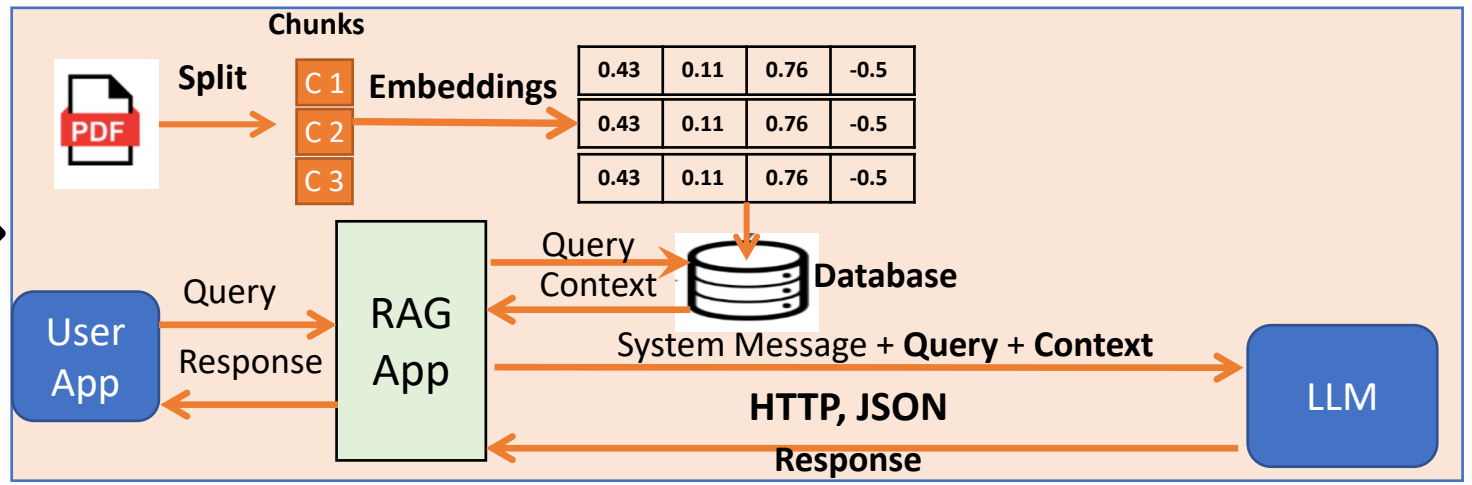
Few Shot Learning

Fine Tuning

RAG
Retrieval
Augmented
Generation







Agentic AI Ecosystem

Agent | Web RTC | Real Time LLM

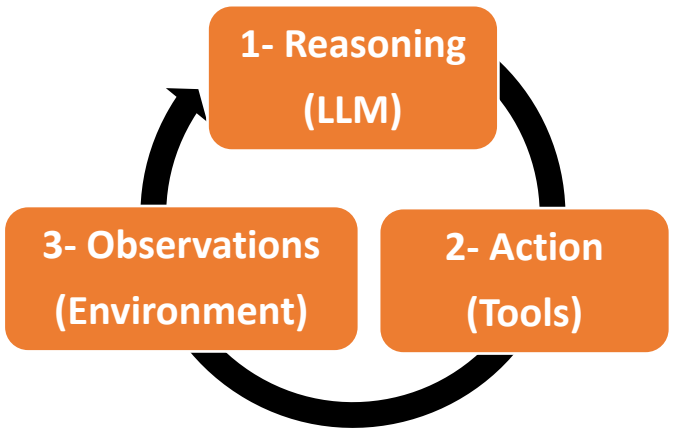
Vous êtes digital Marketer, Je voudrais mettre en vente des voitures dont les caractéristiques sont sont fournies avec :
- Marque, Modèle, Couleur, Puissance fiscale et Prix de vente

Avec ces informations, rédigez une description élégante à propos de ces voitures. Le corps du message doit commencer par un paragraphe très attractif. La description de chaque voiture doit être précédée par un titre contenant la marque et le modèle
Utilisez le format HTML avec des puces pour délimiter les principales fonctionnalités mentionnées dans la description.
Cette description sera envoyée par email à une liste de clients

Merci de faire la compagne de digital marketing

La compagne de digital marketing a été effectuée

Agent : ReaAct (Reasoning/Action) Framework



System message
Prompt | Examples
User Message

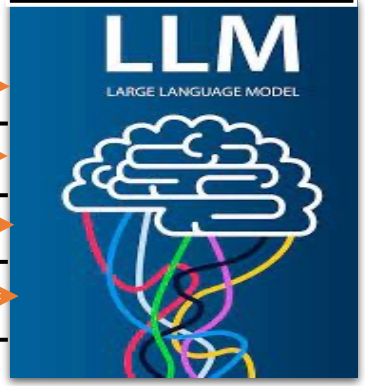
Memory

Stateless Context Window

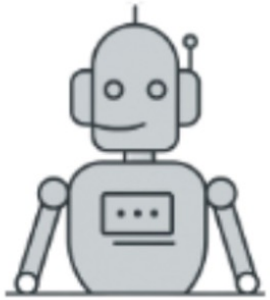
Prompt + Tools + observations

vehicules
customers
sendMail
action 4

Reasoning



- Open AI
- Gemini
- Claude
- Quen
- Deep Seek
- Llamma
- Mistral
- ....



Agent

vehicules | observations

A2A

Agent

MCP | Tools | R4 | R5 | R6

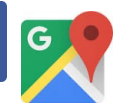
RAG

MCP
Tools
Functions

Busines Layer

Semantic DB

Data base



Action 1 : `get_list_of_selected_vehicules()`

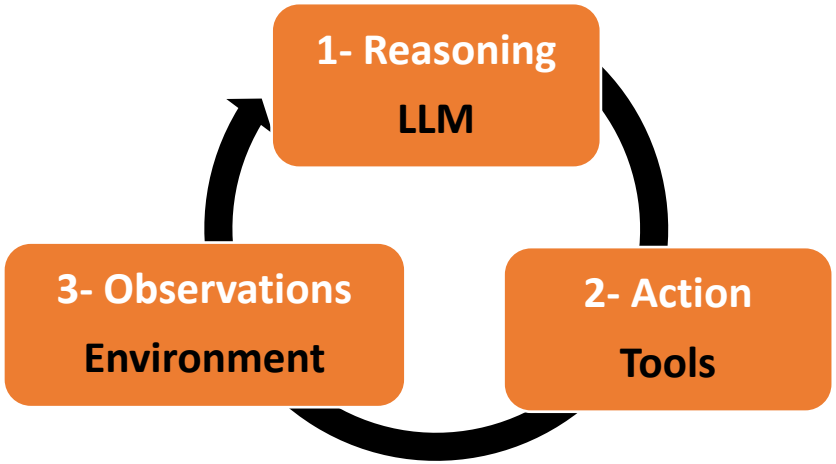
Action 2 : `get_customers()`

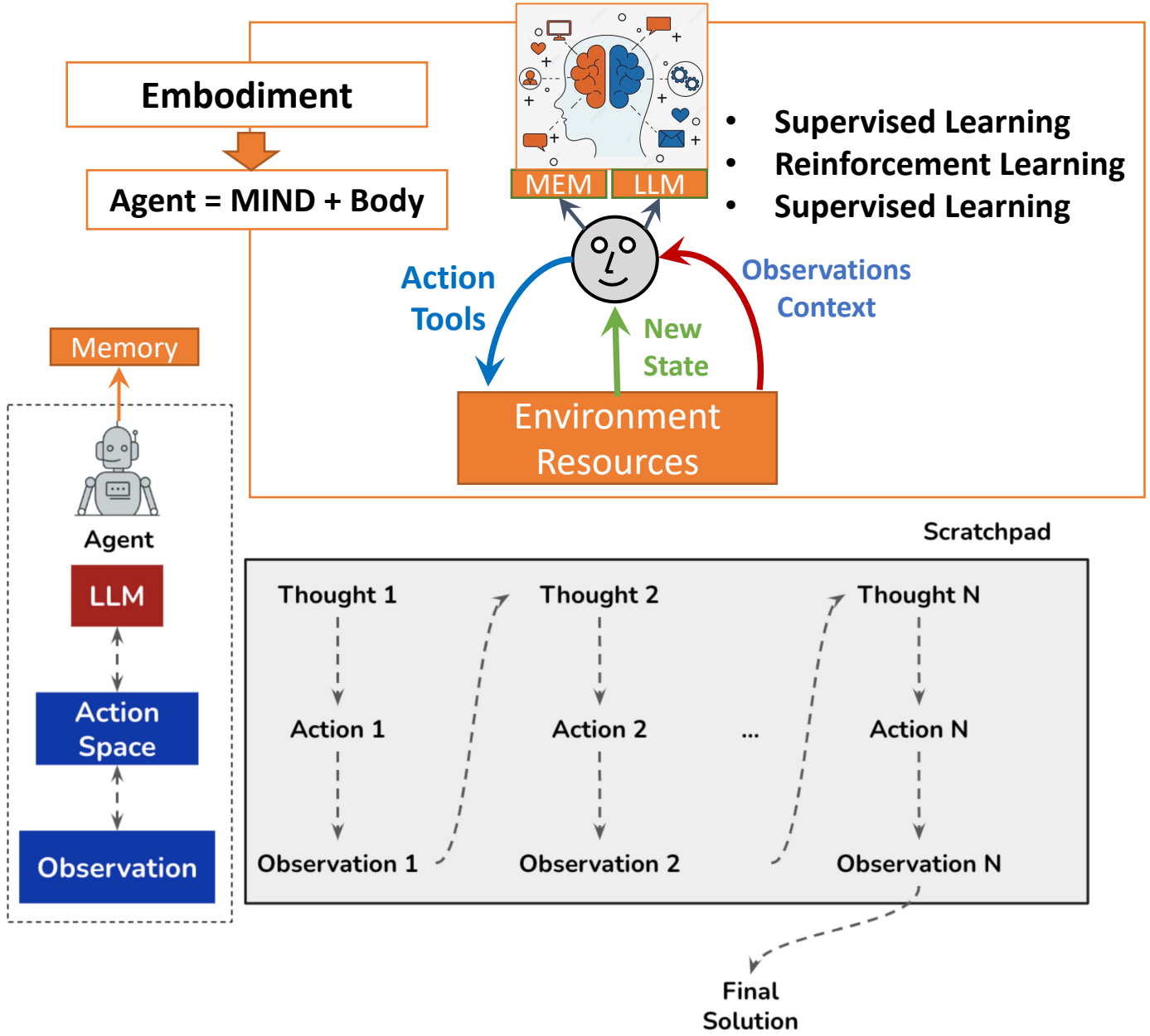
Action 3 : `sendMail(to, subject, content)`

Action 4 : **RAG**

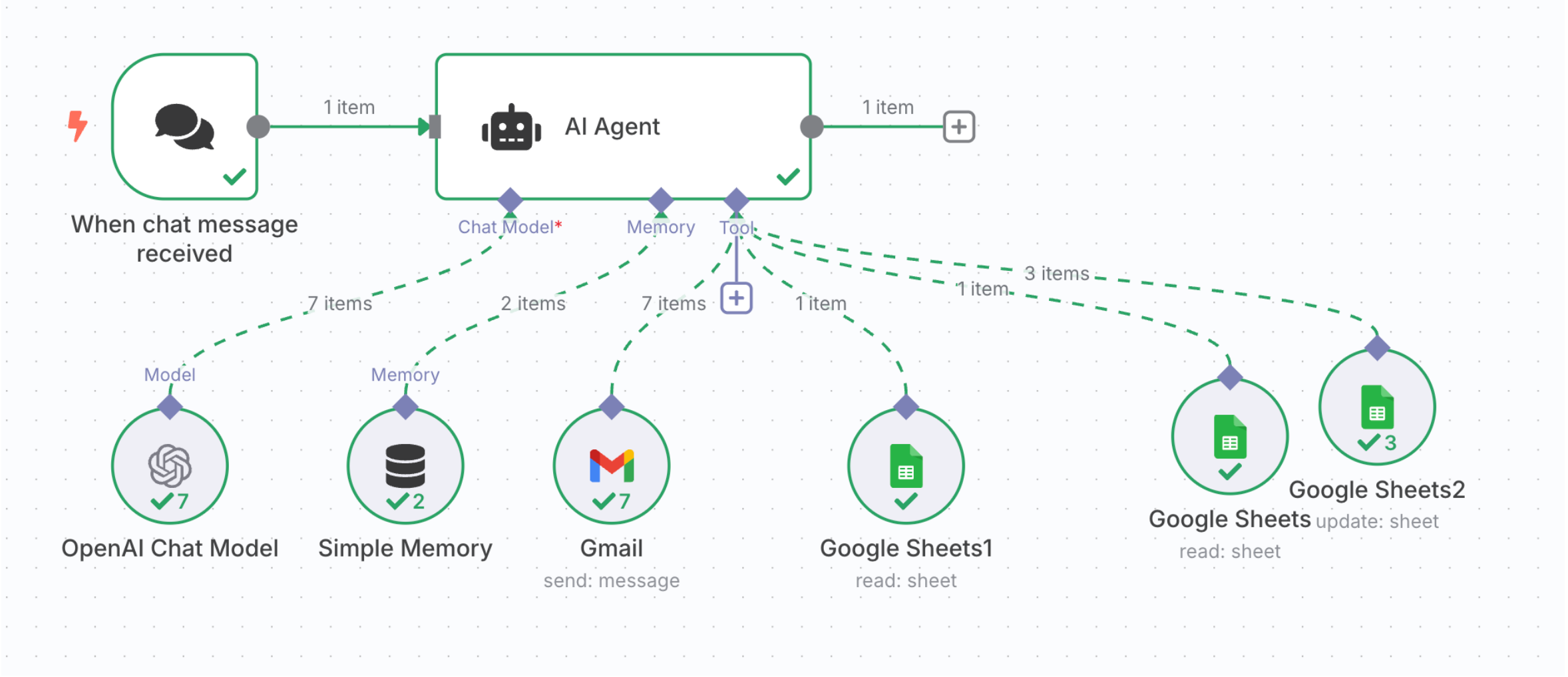
AI Agents

- An AI Agent, as described by the **ReaAct** Framework, is an **autonomous** entity with a goal that can:
  - **Reason** based on context (environmental abstraction)
  - **Act autonomously** on its environment
  - **Collect observations** about the environment





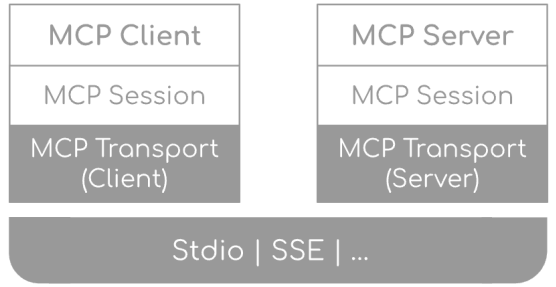
Demo 1 : First Agent in Automation platform : n8n

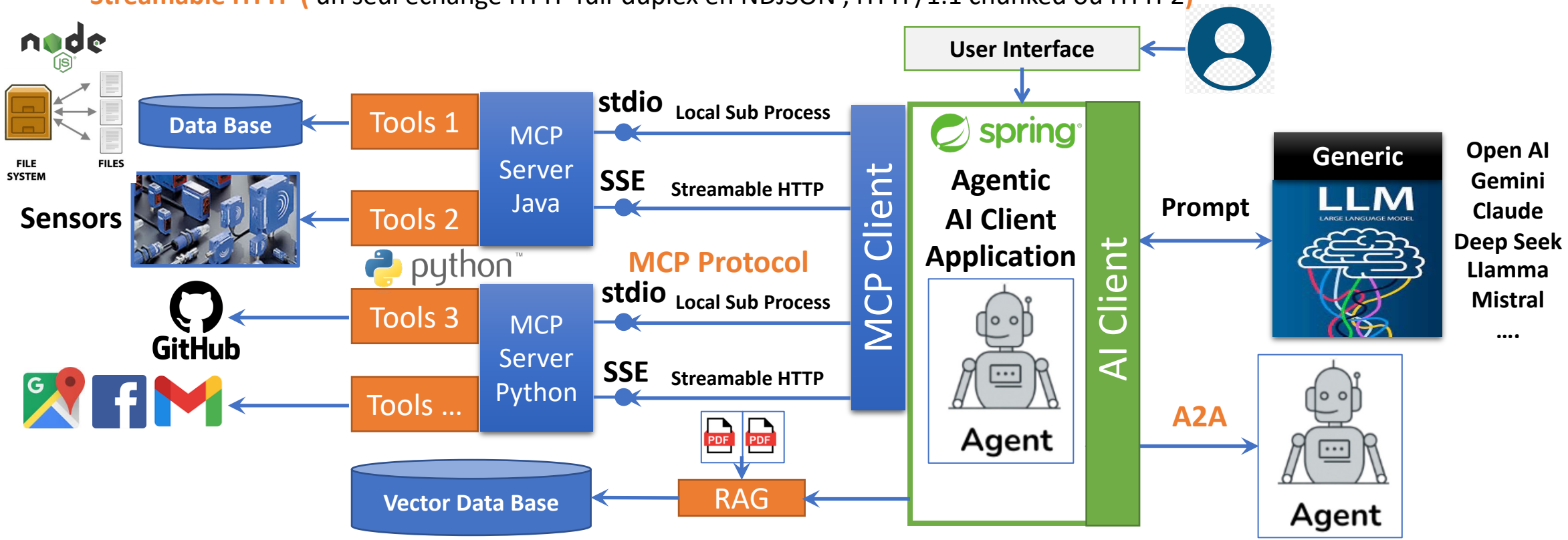


Model Context Protocol (MCP)

https://modelcontextprotocol.io

- MCP standardizes how to integrate **additional context and tools into the ecosystem of AI applications**
- The protocol uses **JSON-RPC 2.0 messages**
- The protocol defines two standard transport mechanisms for client-server communication:
  - **STDIO**, communication over **standard input** and **standard output**
  - **SSE (Server Sent Event :** serveur→client en streaming + POST séparés pour client→serveur
  - **Streamable HTTP (** un seul échange HTTP full-duplex en NDJSON , HTTP/1.1 chunked ou HTTP2**)**





MCP

- Model Context Protocol (MCP) is an open protocol that enables seamless integration between LLM applications and external data sources and tools.
- MCP provides a standardized way for applications to:
  - Share contextual information with language models
  - Expose tools and capabilities to AI systems
  - Build composable integrations and workflows

- The protocol uses **JSON-RPC 2.0 messages** to establish communication between:
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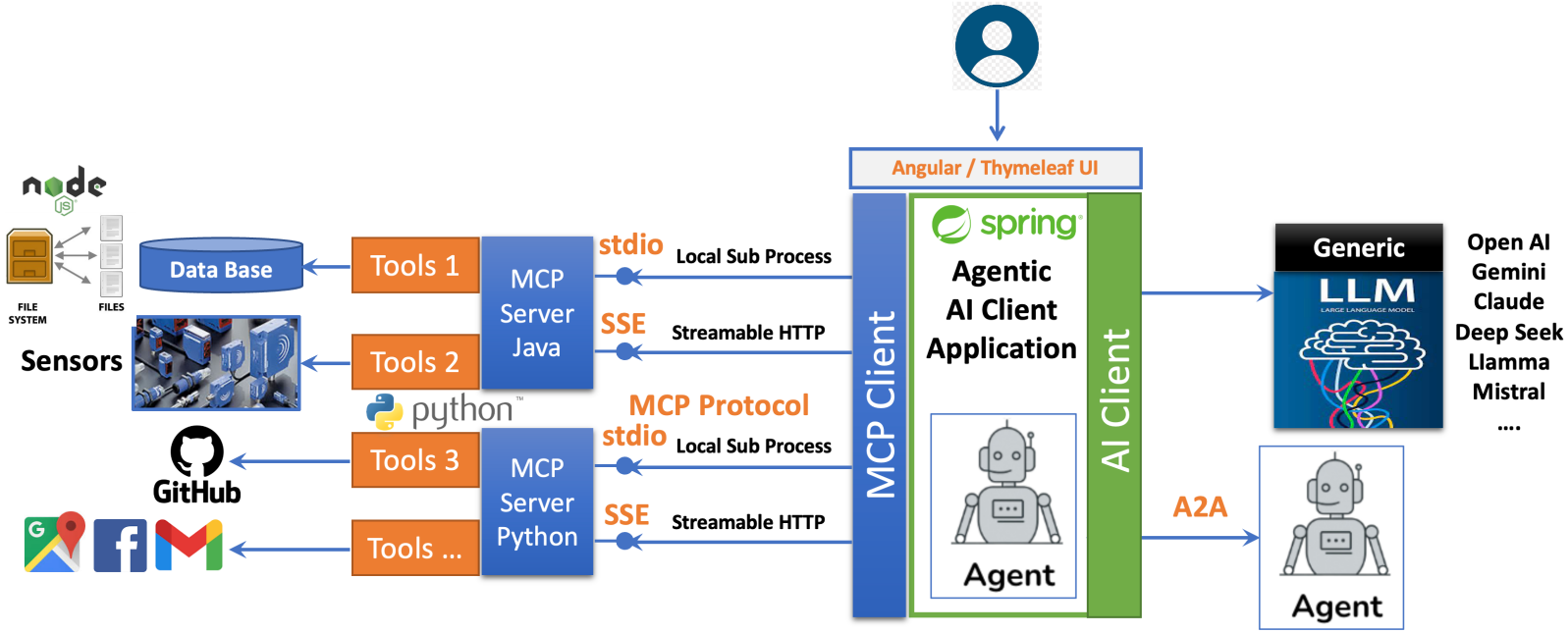
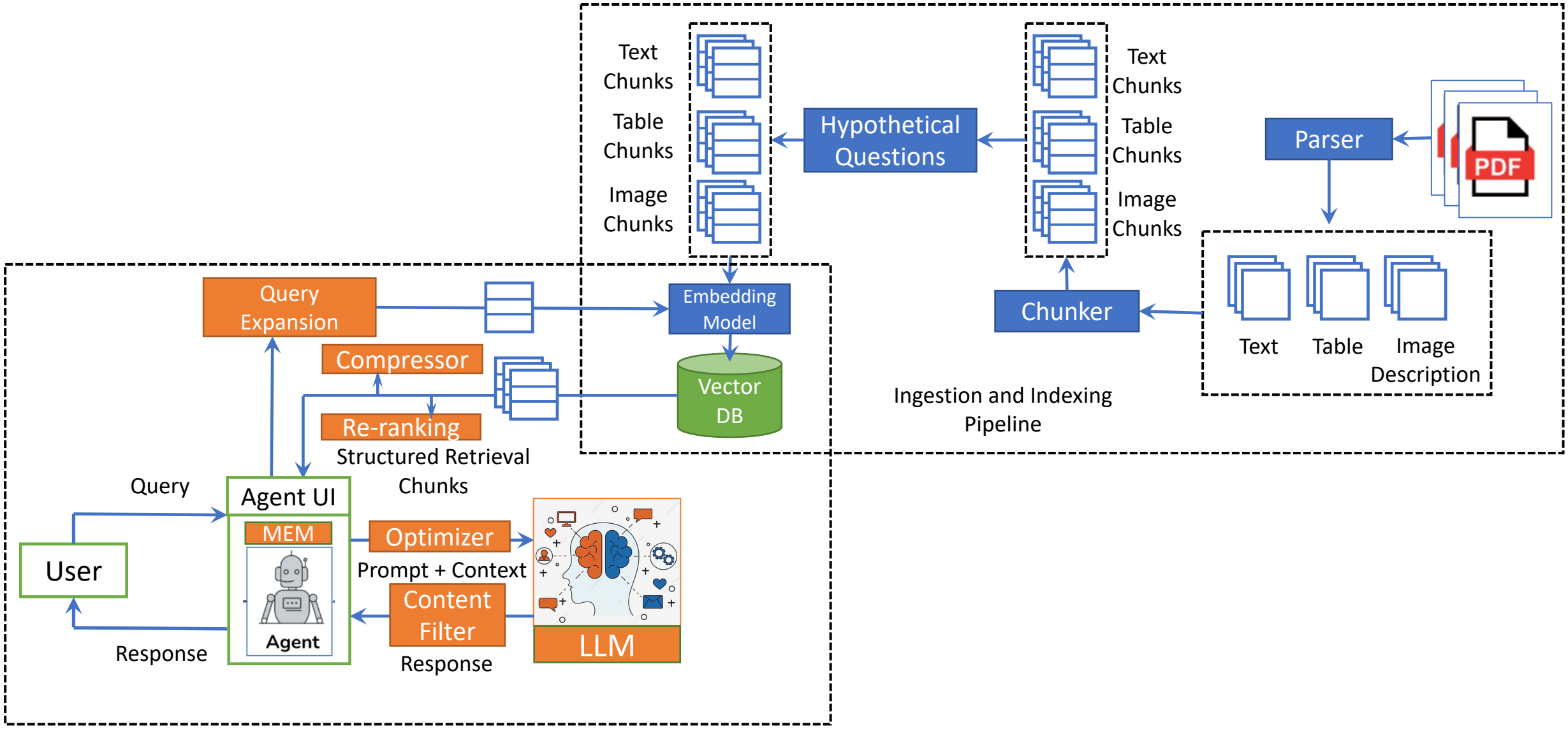
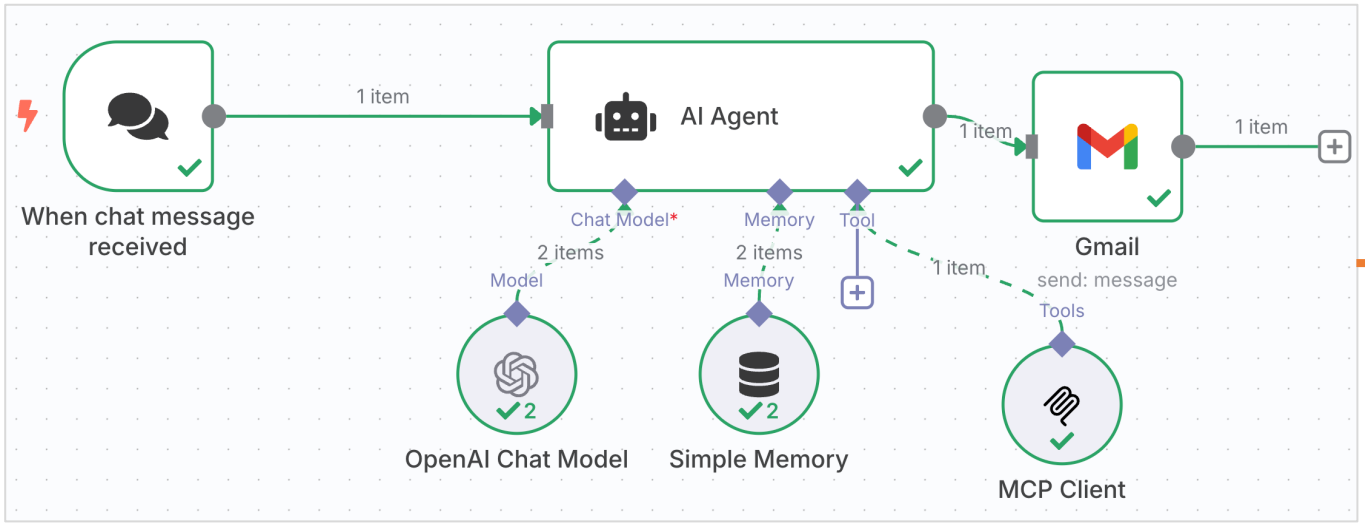


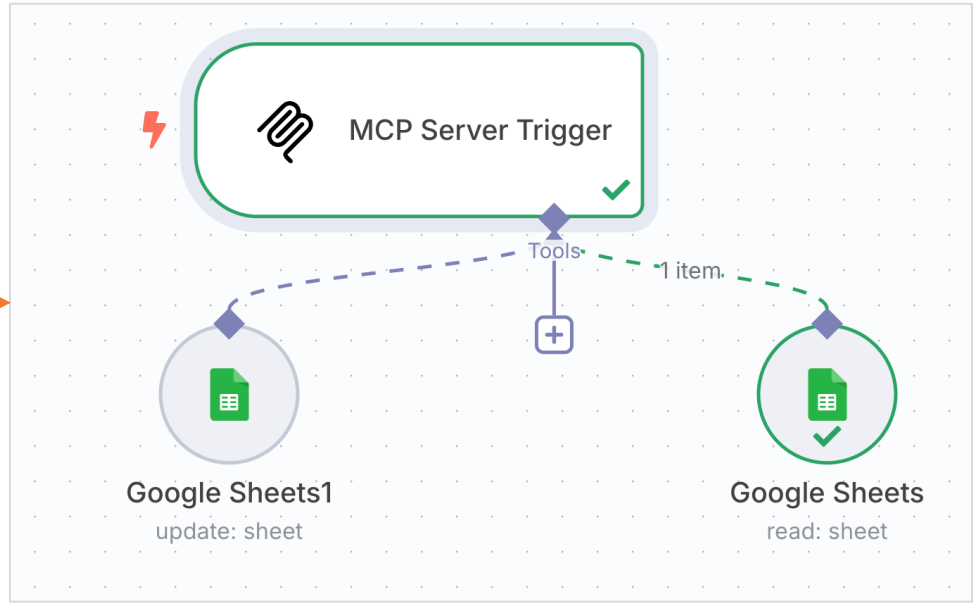
Advanced Multi Modal Agentic RAG



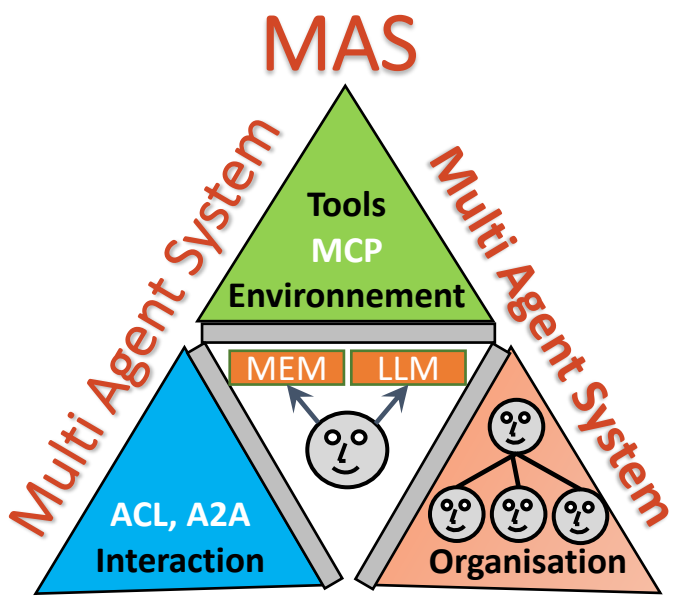
Demo 2 : AI Agent with MCP

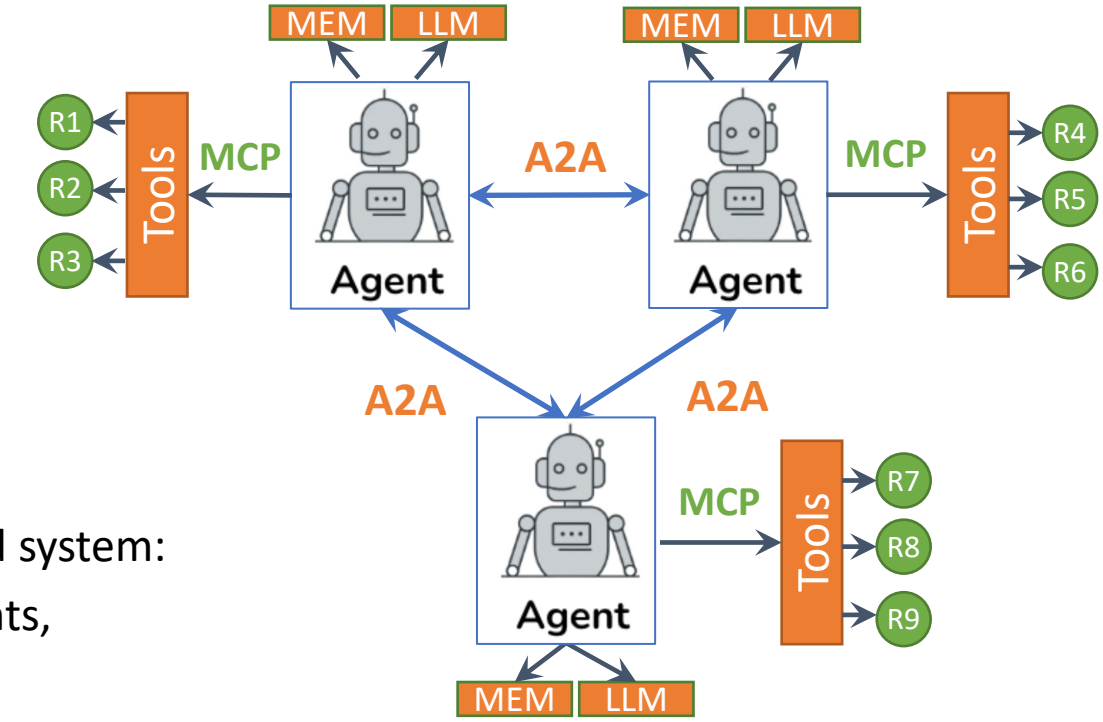


MCP



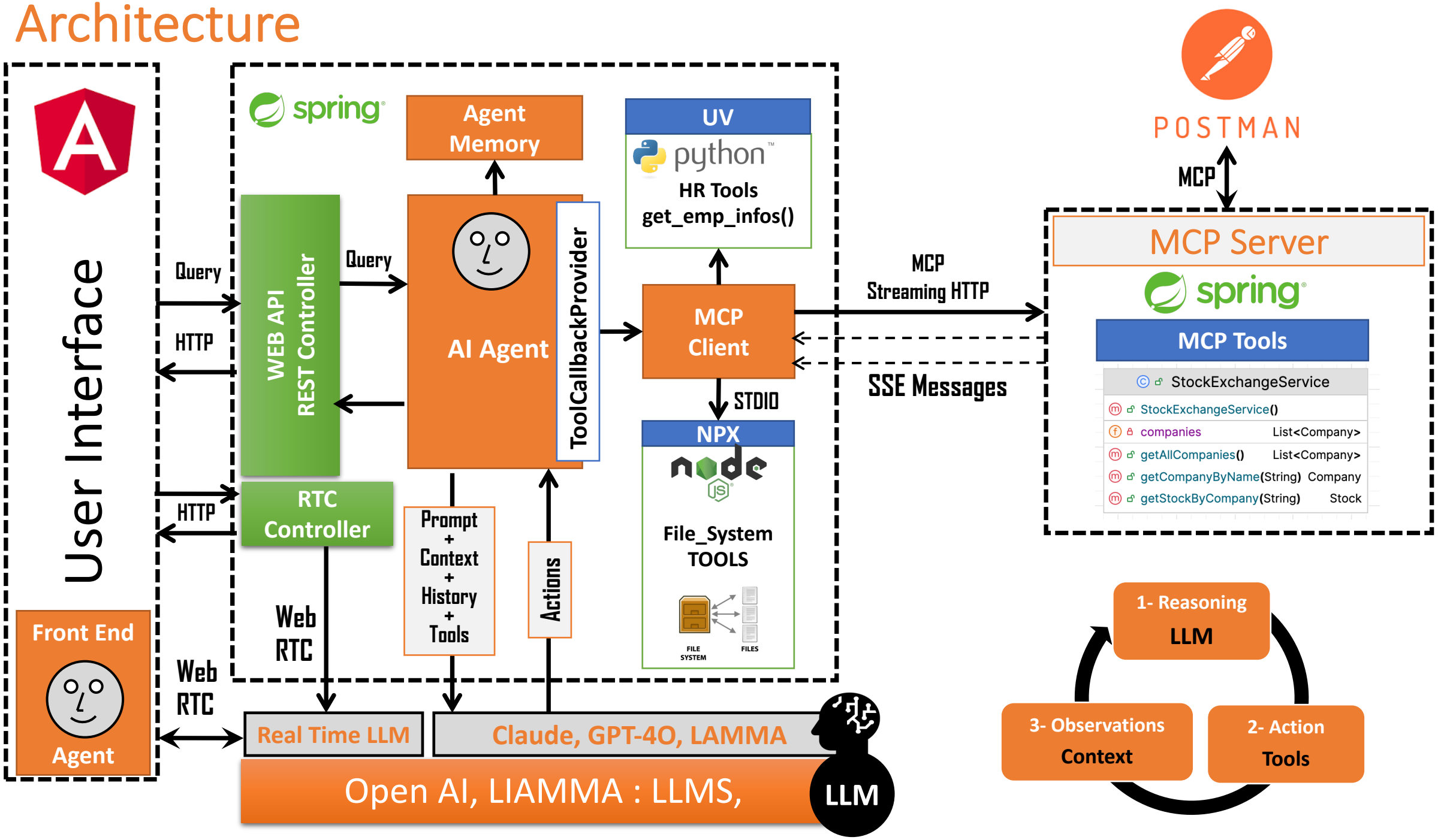
MAS And new Protocols : MCP et A2A



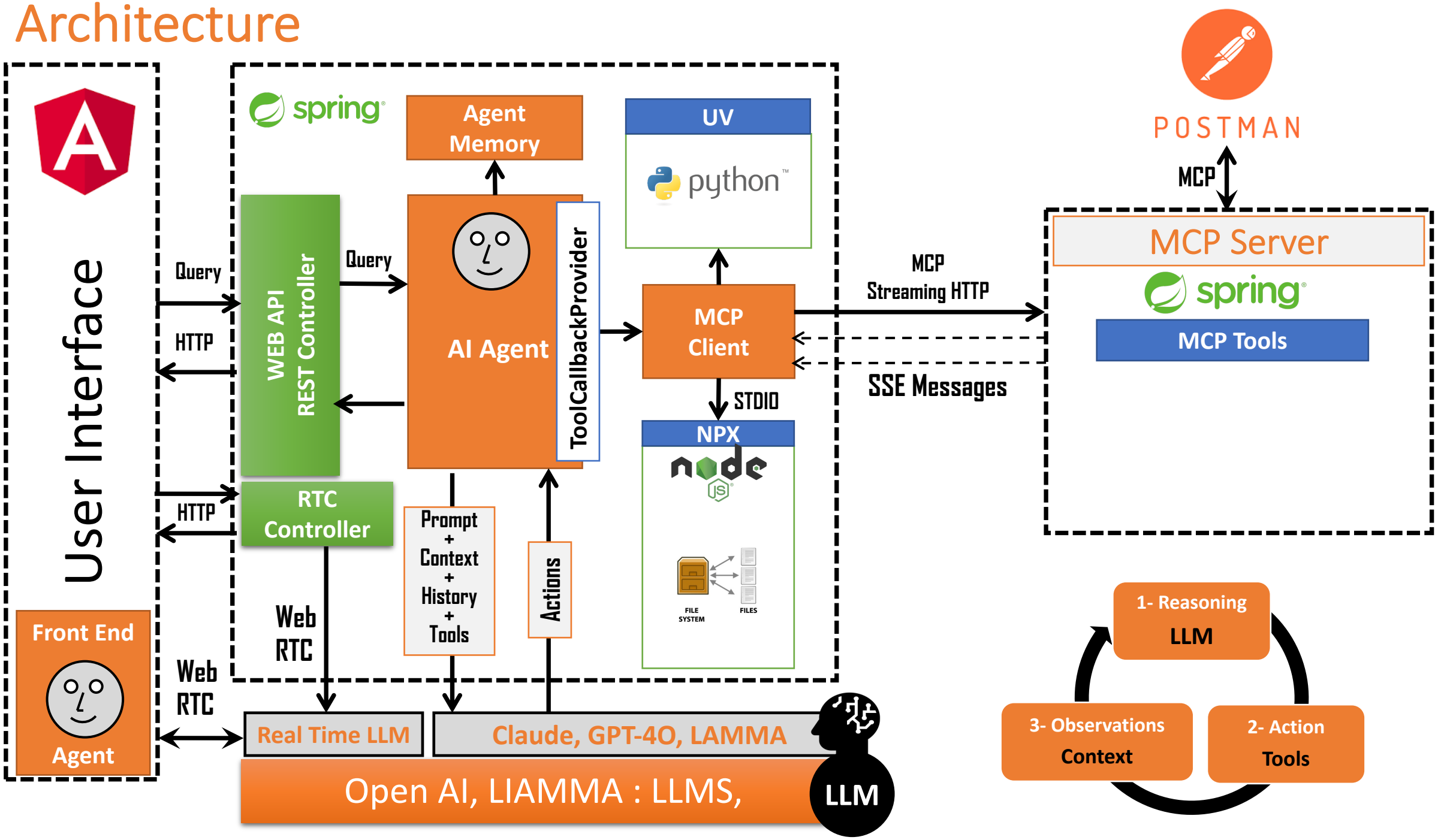


- A multi-agent system (MAS) is a distributed system:
  - Composed of a set of distributed agents,
  - Located in a specific environment
  - and Interacting according to specific organizational structures.
- An MAS solves complex problems by leveraging the collective intelligence of its component agents (DAI).

Architecture



Architecture



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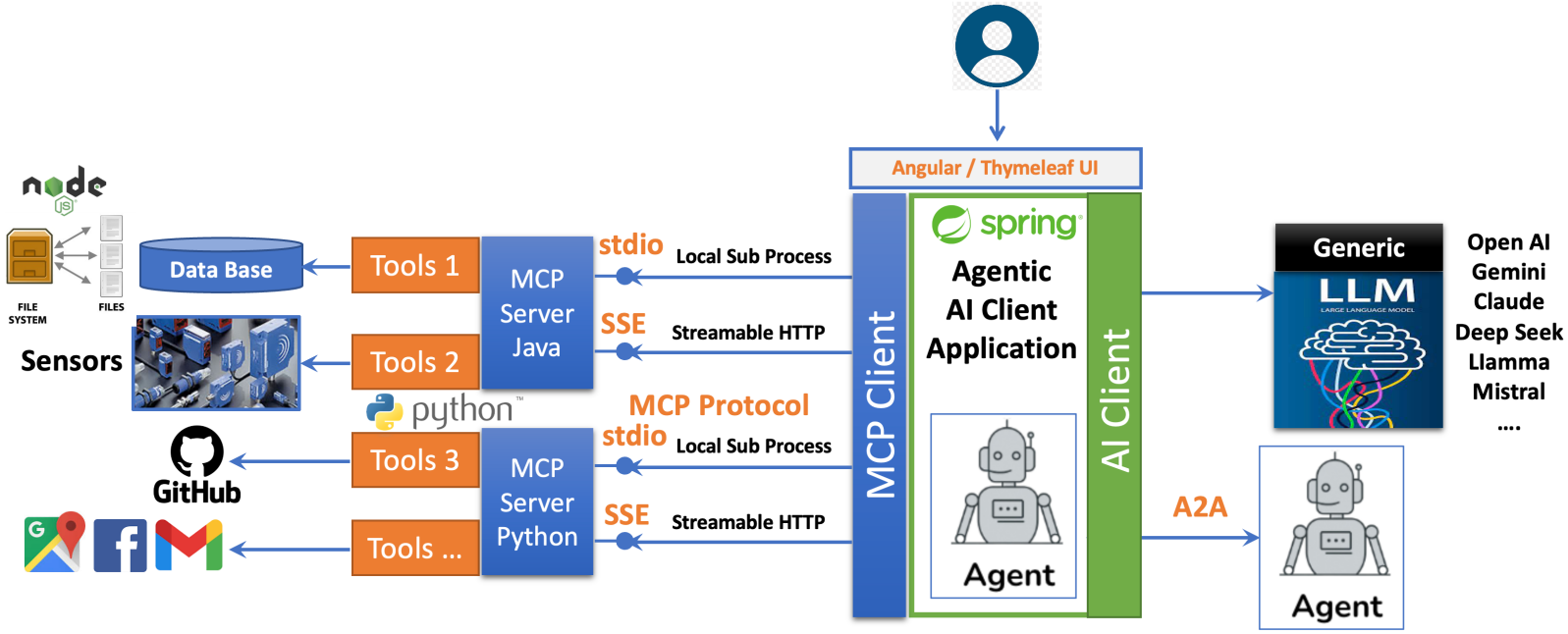
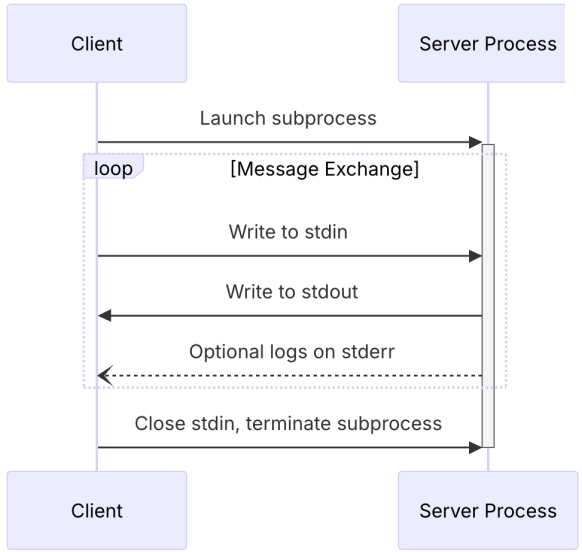
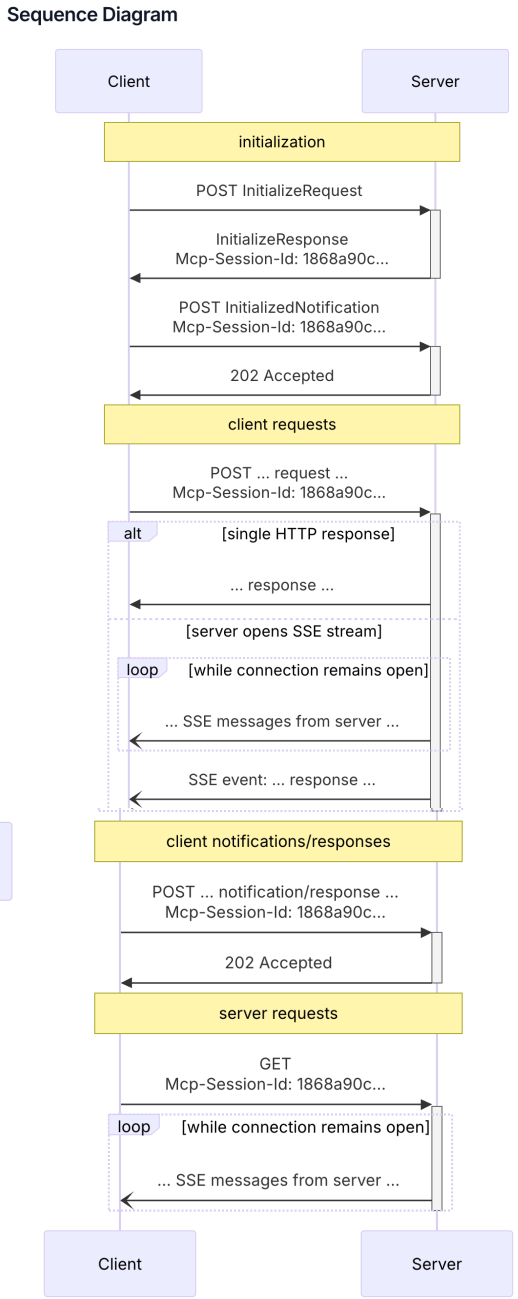


MCP : Transport (stdio)

- The protocol currently defines two standard transport mechanisms for client-server communication:
  - **stdio**, communication over **standard in** and **standard out**
  - **Streamable HTTP (SSE)**
- **STDIO**
  - In the **stdio** transport:
    - The client launches the MCP server as a subprocess.
    - The server reads JSON-RPC messages from its standard input (stdin) and sends messages to its standard output (stdout).
- **Streamable HTTP (SSE)** :
  - In the **Streamable HTTP** transport, the server operates as an independent process that can handle multiple client connections.
  - This transport uses HTTP POST and GET requests.
  - Server can optionally make use of **Server-Sent Events (SSE)** to stream multiple server messages.
  - The server **MUST** provide a single HTTP endpoint path that supports both POST and GET methods. For example, this could be a URL like **https://example.com/mcp**.

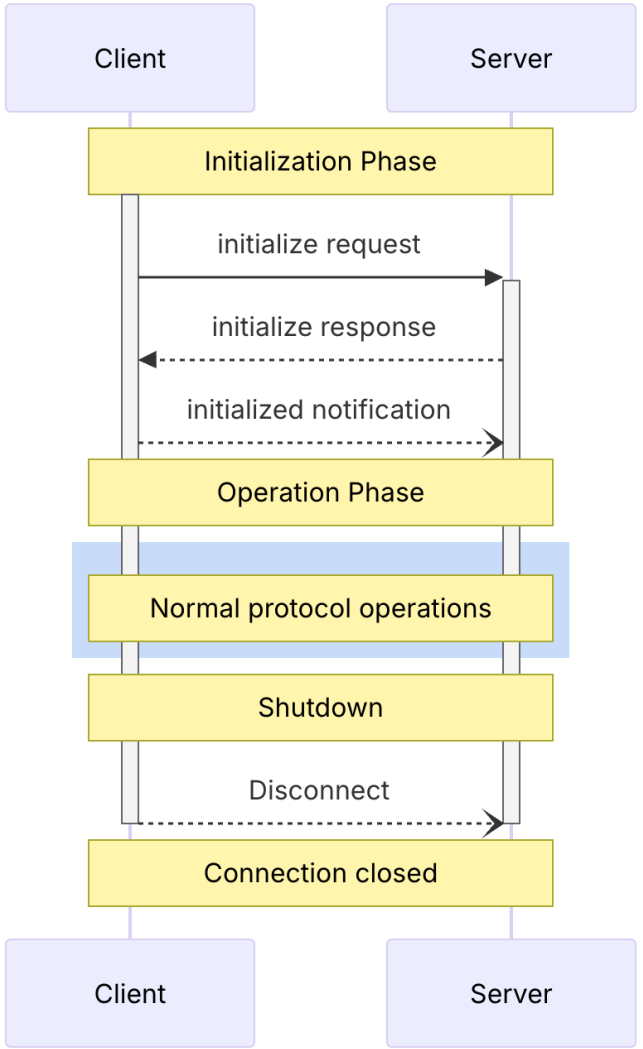




MCP Lifecycle

https://modelcontextprotocol.io/specification/2025-03-26/basic/lifecycle

- **Base Protocol** :
  - JSON-RPC message format
  - Stateful connections
  - Server and client capability negotiation
- **Features :**
  - Servers offer any of the following features to clients:
    - **Resources**: Context and data, for the user or the AI model to use
    - **Prompts**: Templated messages and workflows for users
    - **Tools**: Functions for the AI model to execute
- **MCP Lifecycle:**
  - **Initialization**: Capability negotiation and protocol version agreement
  - **Operation**: Normal protocol communication
  - **Shutdown**: Graceful termination of the connection



MCP Lifecycle : Connection

GET http://localhost:8089/sse Close

Params Authorization Headers (7) Body Scripts Tests Settings Cookies

Query Params

	Key	Value	Description
	Key	Value	Description

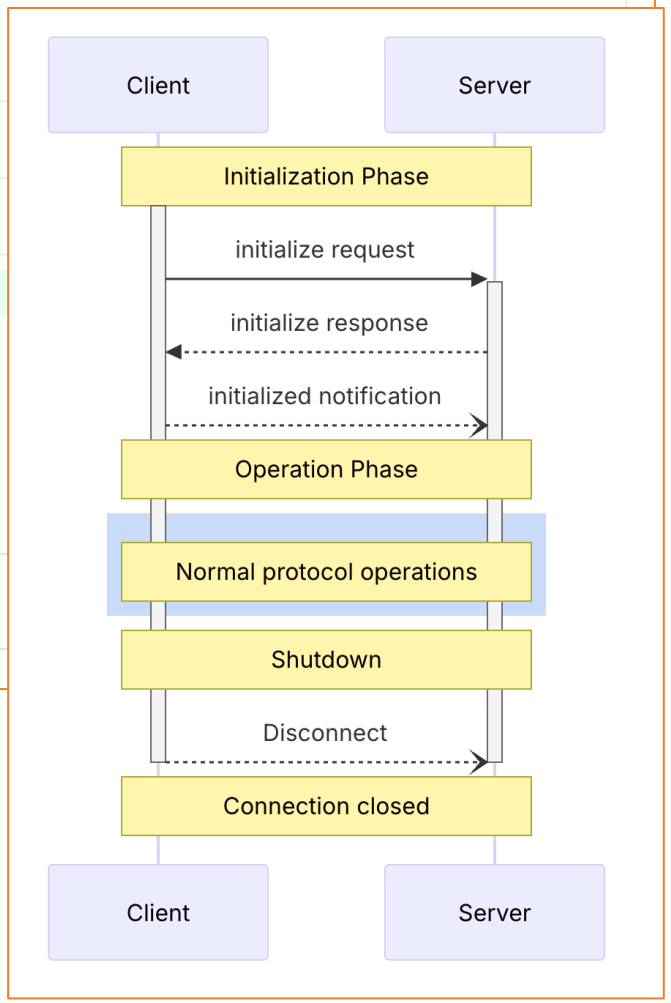
Body Cookies Headers (6) Test Results Receiving Response

Search

endpoint /mcp/message?sessionId=edfa8a68-a3fd-4b53-8cb3-eaa3abc1bd58

Connected to http://localhost:8089/sse

```
$ npx @modelcontextprotocol/inspector
```



MCP Lifecycle : Initialization request

POST | localhost:8089/mcp/message?sessionId=edfa8a68-a3fd-4b53-8cb3-eaa3abc1bd58 | Send

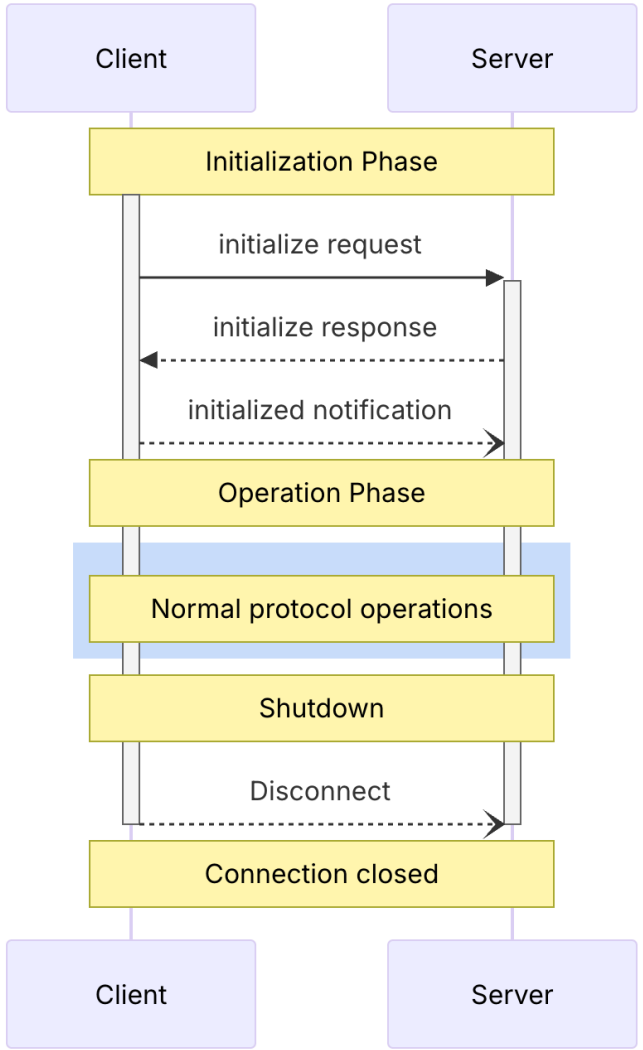
Params • | Authorization | Headers (9) | Body • | Scripts | Tests | Settings

none | form-data | x-www-form-urlencoded | raw | binary | GraphQL | JSON

```json
{
  "jsonrpc": "2.0",
  "id": 1,
  "method": "initialize",
  "params": {
    "protocolVersion": "2024-11-05",
    "capabilities": {
      "roots": {
        "listChanged": true
      },
      "sampling": {}
    },
    "clientInfo": {
      "name": "ExampleClient",
      "version": "1.0.0"
    }
  }
}
```

Server Message Response

```json
{
    "jsonrpc": "2.0",
    "id": 1,
    "result": {
        "protocolVersion": "2024-11-05",
        "capabilities": {
            "completions": {},
            "logging": {},
            "prompts": {
                "listChanged": true
            },
            "resources": {
                "subscribe": false,
                "listChanged": true
            },
            "tools": {
                "listChanged": true
            }
        },
        "serverInfo": {
            "name": "my-mcp-server",
            "version": "1.0.0"
        }
    }
}
```



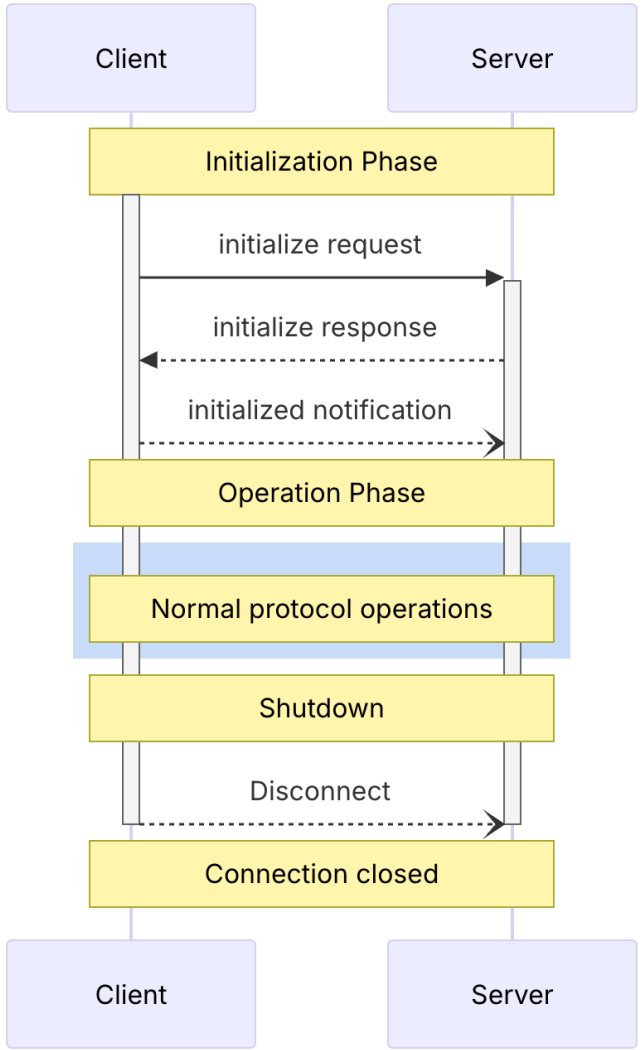
MCP Lifecycle : Initialization notification

POST localhost:8089/mcp/message?sessionId=479a49ec-914f-4be8-9d61-93634e3c6a85 Send

Params • Authorization Headers (9) Body • Scripts Tests Settings

none form-data x-www-form-urlencoded raw binary GraphQL JSON

```
1  {
2  | "jsonrpc": "2.0",
3  | "method": "notifications/initialized"
4  }
```



MCP Lifecycle : Operation phase (tools/list)

POST localhost:8089/mcp/message?sessionId=479a49ec-914f-4be8-9d61-93634e3c6a85 Send

Params • Authorization Headers (9) Body • Scripts Tests Settings

none form-data x-www-form-urlencoded raw binary GraphQL JSON

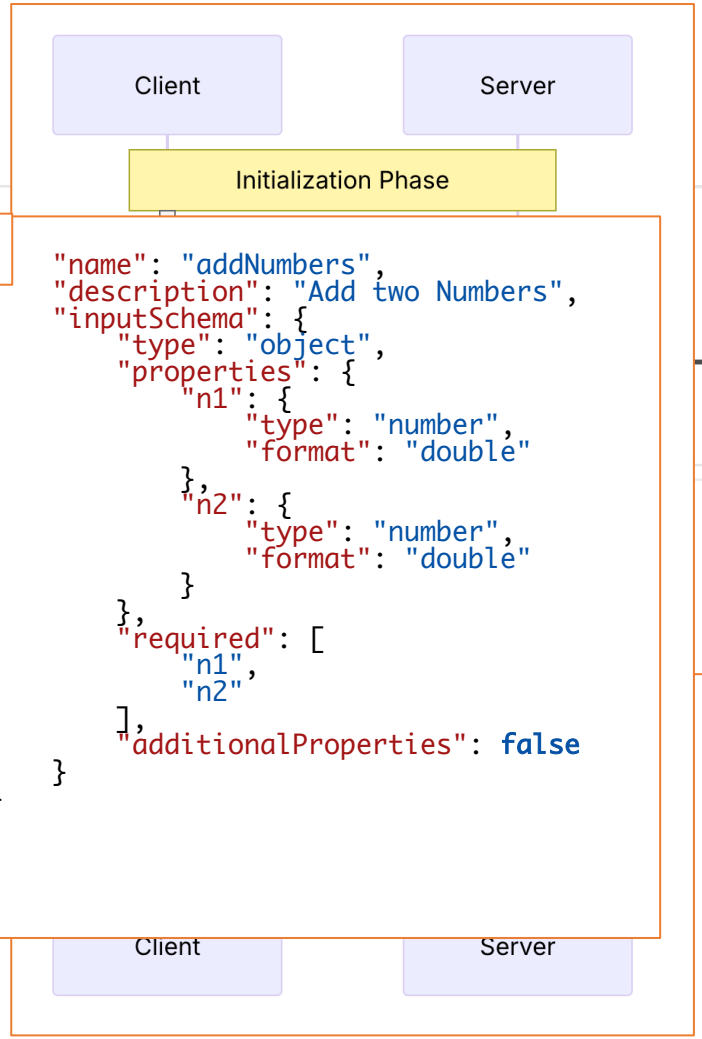
```
{
  "jsonrpc": "2.0",
  "method": "tools/list",
  "id":1
}
```

Body Cookies Headers (4) Test Results

Raw Preview Visualize

Server Message Response


```
{
    "jsonrpc": "2.0",
    "id": 1,
    "result": {
        "tools": [
            {
                "name": "multiplyNumbers",
                "description": "Multiply two Numbers",
                "inputSchema": {
                    "type": "object",
                    "properties": {
                        "n1": {
                            "type": "number",
                            "format": "double"
                        },
                        "n2": {
                            "type": "number",
                            "format": "double"
                        }
                    },
                    "required": [
                        "n1",
                        "n2"
                    ],
                    "additionalProperties": false
                }
            },
            {
                "name": "addNumbers",
                "description": "Add two Numbers",
                "inputSchema": {
                    "type": "object",
                    "properties": {
                        "n1": {
                            "type": "number",
                            "format": "double"
                        },
                        "n2": {
                            "type": "number",
                            "format": "double"
                        }
                    },
                    "required": [
                        "n1",
                        "n2"
                    ],
                    "additionalProperties": false
                }
            }
        ]
    }
}
```

MCP Lifecycle : Operation phase (tools/call)

POST localhost:8089/mcp/message?sessionId=3c907428-fb16-465e-8b27-a615f658d34d **Send**

Params • Authorization Headers (9) Body • Scripts Tests Settings

none form-data x-www-form-urlencoded raw binary GraphQL JSON

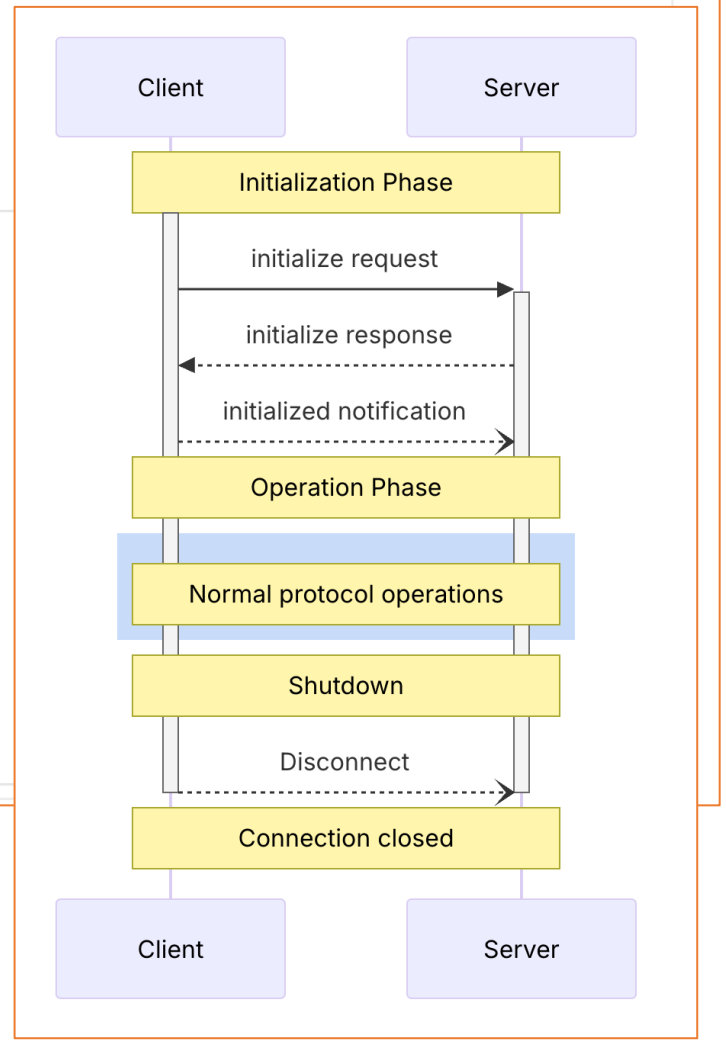
```
{
  "jsonrpc": "2.0",
  "method": "tools/call",
  "params":{
    "name":"addNumbers",
    "arguments":{
        "n1":34,
        "n2":11
    }
  },
  "id":1
}
```

Server Message Response

message {"jsonrpc":"2.0","id":1,"res

JSON

```
{
    "jsonrpc": "2.0",
    "id": 1,
    "result": {
        "content": [
            {
                "type": "text",
                "text": "45.0"
            }
        ],
        "isError": false
    }
}
```



MCP Lifecycle : Operation phase (tools/call

Untitled MCP Request

Save | Share

HTTP | localhost:8089/sse | Run

Message | Authorization | Headers | Settings

Tools | Prompts | Resources

multiplyNumbers

Multiply two Numbers

n1* : number

78

n2* : number

12

2. addNumbers

Add two Numbers

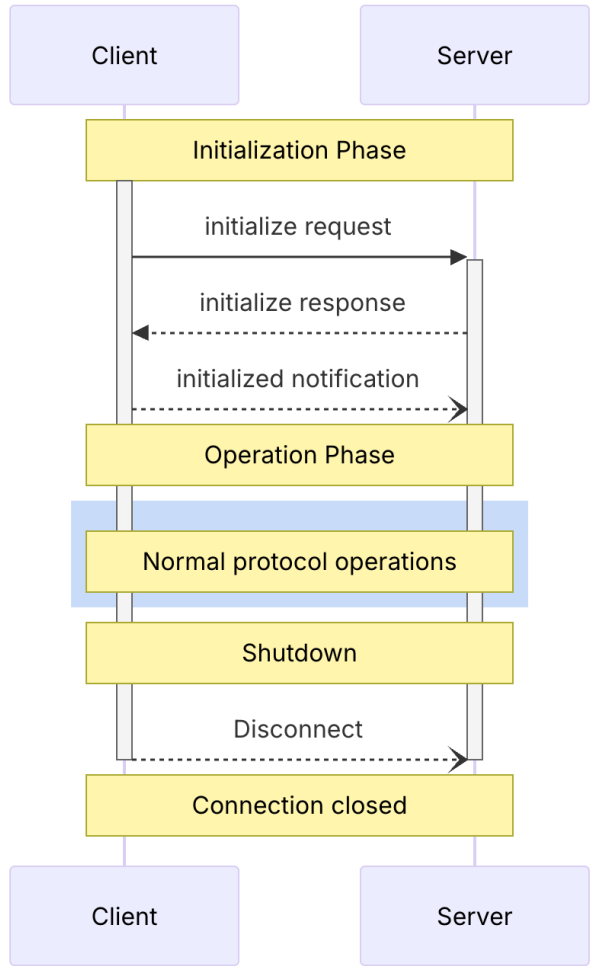
Response | Notifications

Search | All Messages

Connected 23:12:16.076

Server Message Response

```
{
  "method": "tools/call",
  "params": {
    "name": "multiplyNumbers",
    "arguments": {
      "n1": 78,
      "n2": 12
    }
  }
}
```



AI Agents Frameworks

- **OpenAI Agent SDK** : https://openai.github.io/openai-agents-python/
- **CrewAI** : https://docs.crewai.com/introduction
- **LangChain** : https://python.langchain.com/docs/tutorials/agents/
- **AutoGen** : https://microsoft.github.io/autogen/dev//user-guide/agentchat-user-guide/tutorial/agents.html
- **Pydantic AI Agent** : https://ai.pydantic.dev/
- **Phidata** : https://docs.phidata.com/introduction
- **PraisonAiAgents** : https://docs.praison.ai/framework/praisonaiagents
- …

OpenAI Agents SDK

- The OpenAI Agents SDK aims to build agentic AI apps in a lightweight, easy-to-use package with very few abstractions.
- It's a production-ready upgrade of our previous experimentation for agents, Swarm.
- The Agents SDK has a very small set of primitives:
  - **Agents**, which are LLMs equipped with instructions and tools
  - **Handoffs**, which allow agents to delegate to other agents for specific tasks
  - **Guardrails**, which enable the inputs to agents to be validated
  - Support **Model Context Protocol (MCP)**

Installation

```
pip install openai-agents
```

Hello world example

```
from agents import Agent, Runner

agent = Agent(name="Assistant", instructions="You are a helpful assistant")

result = Runner.run_sync(agent, "Write a haiku about recursion in programming.")
print(result.final_output)

# Code within the code,
# Functions calling themselves,
# Infinite loop's dance.
```

```
export OPENAI_API_KEY=sk-...
```

DEMO : Agentic AI

Source code and resources : https://github.com/mohamedYoussfi/agentic-ai

```
> pip install uv
> uv init
> uv venv
> source .venv/bin/activate (MACOS)
> .venv\Scripts\activate.bat (Windows)
> uv add ipykernel
> uv add "openai-agents[viz]"
> uv add pydantic
> uv add python-dotenv
> uv add streamlit
```

- **pyproject.toml**

```
[project]
name = "test"
version = "0.1.0"
description = "Add your description here"
readme = "README.md"
requires-python = ">=3.11"
dependencies = [
    "duckduckgo-search>=8.0.1",
    "ipykernel>=6.29.5",
    "openai-agents[viz]>=0.0.14",
    "pydantic>=2.11.4",
    "python-dotenv>=1.1.0",
    "streamlit>=1.45.0",
]
```

First Agent Using OpenAI Model

```
from dotenv import load_dotenv
import os
from IPython.display import Markdown
```

✓ 0.0s

```
load_dotenv()
```

✓ 0.0s

```
api_key = os.environ.get('OPENAI_API_KEY')
if not api_key:
    raise("API KEY not available")
```

✓ 0.0s

.env

```
OPENAI_API_KEY=sk-proj-…
```

```
from agents import Agent, Runner, OpenAIChatCompletionsModel, AsyncOpenAI
```

✓ 0.5s

```
cost_medico_agent = Agent(
    name="Doctor Agent",
    model="gpt-4o",
    instructions="""
    Donne moi des conseils concernant la pathologie fournie par l'utilisateur
    """
    )
result =  await(Runner.run(cost_medico_agent,"Cardiologie"))
```

```
display(Markdown(result.final_output))
```

En ce qui concerne la cardiologie, il est important de prendre soin de la santé cardiovasculaire en suivant plusieurs conseils généraux:

1. **Alimentation équilibrée** : Adopte un régime riche en fruits, légumes, grains entiers, protéines maigres et acides gras oméga-3 (présents dans le poisson, les noix et les graines).
2. **Activité physique régulière** : Vise au moins 150 minutes d'exercice modéré par semaine, comme la marche rapide, la natation ou le vélo.
3. **Gestion du stress** : Pratiques comme le yoga, la méditation ou des exercices de respiration peuvent aider à réduire le stress.
4. **Arrêt du tabac** : Si tu fumes, chercher de l'aide pour arrêter peut drastiquement améliorer ta santé cardiaque.
5. **Contrôle de la pression artérielle et du cholestérol** : Effectue des contrôles réguliers et prends les mesures nécessaires si les niveaux sont élevés.
6. **Maintien d'un poids sain** : Un poids adéquat peut réduire le risque de maladies cardiovasculaires.
7. **Modération de la consommation d'alcool** : Limite ta consommation d'alcool si tu choisis de boire.
8. **Suivi médical** : Effectue des bilans de santé réguliers pour détecter précocement les problèmes.

Si tu as des soucis spécifiques ou un diagnostic, il est important de consulter un cardiologue pour un plan de traitement adapté à ton cas.

First Agent Using OpenAI Llamma 3.2 Model

```python
lamma_model = OpenAIChatCompletionsModel(
    model="llama3.2",
    openai_client= AsyncOpenAI(base_url="http://localhost:11434/v1")
)
```

```python
free_medico_agent = Agent(
    name="Doctor Agent",
    model=lamma_model,
    instructions="""
    Donne moi des conseils concernant la pathologie fournie par l'utilisateur
    """
    )
result =  await(Runner.run(free_medico_agent,"Cardiologie"))
```

```python
display(Markdown(result.final_output))
```

Voici quelques informations générales et conseils relatifs à la cardiologie :

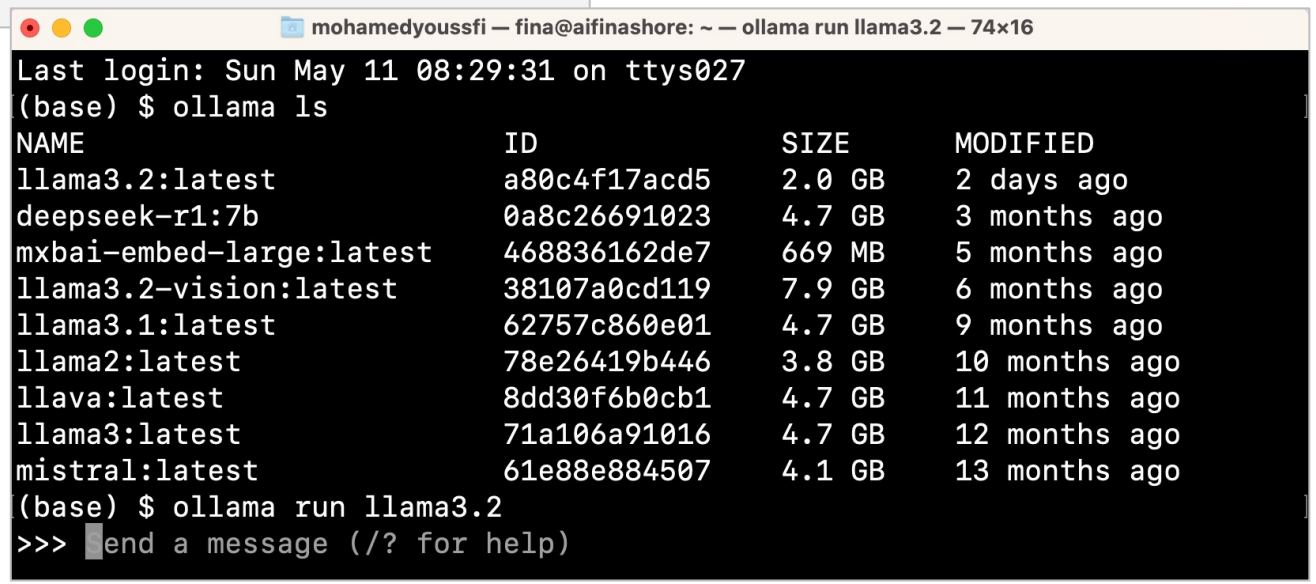
Qu'est-ce que le cœur ?

Le cœur est un organe essentiel de notre corps qui frappe sur environ 100 000 fois par jour. Il est responsable de pousser l' oxygène à travers les artères et d'amener à retourner au cœur le déchet, comme les déchets du cerveau. Le cœur est établi à la base du torso.

Les troubles cardiaques

Il existe une large gamme de maladies cardiovasculaires qui affectent millions de personnes dans le monde entier. Voici quelques exemples :

1. **Insuffisance cardiaque**: Lorsqu'elles ne fonctionnent pas avec suffisamment pour fournir les doses nécessaires à tout le corps, les organes peuvent s'empirer.
2. **Coronaropathie** : Cette maladie est due au stress des artères. Tout ce que vous avez besoin de faire, c'est que votre système d'ongulates réagisse très bien avec une pression.
3. **Aortique et pulmonaire hypertension**: le risque de stent ou d'assister à un cedran
4. **Arérodite hémorragique** : sache qu'il y a peut-être des problèmes en cas d'hémorragie
5. **Anomalies du coeur (AESC)**



Agent and Structured Output

```python
from pydantic import BaseModel
```

```python
class Recipe(BaseModel):
    title : str
    ingredients : list[str]
    cooking_time : int
    servings : int
```

```python
recipe_agent = Agent(
    name = "Recipe Agent",
    model="gpt-4o",
    instructions= """
    You are an agent for creating Recipes
    You will be given the name of the food to create
    You job is to create this food as an actual detailed recipe
    The cooking time should be in minutes
    """,
    output_type= Recipe
)
```

```python
response = await Runner.run(recipe_agent,"Tajine Marocain en Arabe")
recipe = response.final_output
```

```python
print("*"*40)
print(f"Title : {recipe.title}")
print(f"Time : {recipe.cooking_time} minutes")
print(f"Servings : {recipe.servings}")
for ing in recipe.ingredients:
    print(f"- {ing}")
```

```
****************************************
Title : طاجين مغربي
Time : 120 minutes
Servings : 4
- 1 كغ من لحم الغنم (أو الدجاج حسب الرغبة)
- 2 بصلات متوسطة الحجم، مقطعة إلى شرائح
- 3 فصوص ثوم مفرومة
- 4 ملاعق كبيرة من زيت الزيتون
- 1 ملعقة صغيرة من الزنجبيل المطحون
- 1 ملعقة صغيرة من الكركم
- 1 ملعقة صغيرة من القرفة
- ملح وفلفل حسب الرغبة
- 100 غرام من الزبيب
- 100 غرام من اللوز المقلي
- عصير نصف ليمونة
- 2 كوب من الماء
- حفنة من الكزبرة الطازجة، مفرومة للزينة
- 200 غرام من الخضروات الموسمية (مثل الجزر والبطاطس)، مقطعة
```

Agent and Tools

```
response = await Runner.run(weather_agent, "The temperature and weather in Marrakech")
response.final_output
```

```
The tool get_weather is used by the agent to get the weather of the Marrakech
The tool get_temperature is used by the agent to get the temperature of the Ma

'The weather in Marrakech is sunny, with a temperature of 50°F.'
```

```
from agents import Agent, function_tool, WebSearchTool
```

```
@function_tool
def get_weather(city : str) -> str:
    """
    Get the weather of a given city
    """
    print(f"The tool get_weather is used by the agent to get the weather of the {city}")
    return f"The weather of {city} is sunny"
```

```
response = await Runner.run(weather_agent, "Quel temps fera t-il demain à Tanger")
response.final_output
```

```
The tool get_weather is used by the agent to get the weather of the Tanger

'Demain à Tanger, le temps sera ensoleillé. ☀️'
```

```
@function_tool
def get_temperature(city : str) -> str:
    """
    Get the teperature in a given city
    """
    print(f"The tool get_temperature is used by the agent to get the temperature of the {city}")
    return f"it is 50° In {city}"
```

```
weather_agent = Agent (
    name="The Weather Agent",
    instructions="""
                As the local weather agent, you will be asked to give information
                about weather of a given city,
                Your output should include weather information asked by the user"
                """,
    tools=[get_weather, get_temperature]
)
```

Agent and Tools

```
search_agent = Agent(
    name= "Search Agent",
    instructions= """
    You are a news reporter,
    your job is to search new recent articles using internet about a given country
    """,
    tools=[WebSearchTool()]
)
```

```
search_agent = Agent(
    name= "Search Agent",
    instructions= """
    You are a news reporter,
    your job is to search new recent articles using internet about a given country
    """,
    tools=[get_news_articles]
    )
```

```
from duckduckgo_search import DDGS
from datetime import datetime
```

```
now = datetime.now().strftime("%Y-%m-%d")
now
```

```
@function_tool
def get_news_articles(topic:str):
    print(f"Running DuckDuckSearch for news in topic : {topic}")
    ddgs_api = DDGS()
    results = ddgs_api.text(f"{topic}, {now}", max_results=5)
    return results
```

```
response = await Runner.run(search_agent, "Les actualités au Maroc")
```

```
display(Markdown(response.final_output))
```

```
Running DuckDuckSearch for news in topic : المغرب
```

```
display(Markdown(response.final_output))
```

✓ 0.0s

آخر الأخبار حول المغرب

1. **حكومة بيدرو سانشيز تمنح المغرب 340 مليون يورو لبناء مشروع محطة تحلية المياه**
   - الرابط: اقرأ المزيد
   - مقتطف: حكومة بيدرو سانشيز منحت المغرب مبلغ 340 مليون يورو لتمويل مشروع محطة لتحلية المياه...
2. **التقويم والأعياد الرسمية للمغرب سنة 2025**
   - الرابط: اقرأ المزيد
   - مقتطف: تشمل العطل الرسمية في المغرب عام 2025 رأس السنة الجديدة وذكرى بيان الاستقلال...
3. **المملكة المغربية لسنة 2025 - تفاصيل العطل**

Agent and Handoffs

```python
from IPython.display import Markdown
from agents.extensions.visualization import draw_graph
```

```python
from agents import Agent
from pydantic import BaseModel
```

```python
class Tutorial(BaseModel):
    outline : str
    tutorial : str
```

```python
tutorial_response = await Runner.run(outline_builder_agent,
                                     "Boucles en Java avec des explications en Français")
tuto = tutorial_response.final_output
```

```python
tutorial_generator_agent = Agent(
    name="Tutorial Generator Agent",
    handoff_description= "Used to generate tutorial for a given outline",
    instructions="""
        Given a programming topic and an outline,
        your job is to generate code snippets for each section of the outline,
        Format the tutorial in Markdown using a mix of text explaination and code snippets examples",
        Include comments in the code snippets to further explain the code"
    """,
    output_type=Tutorial
)
```

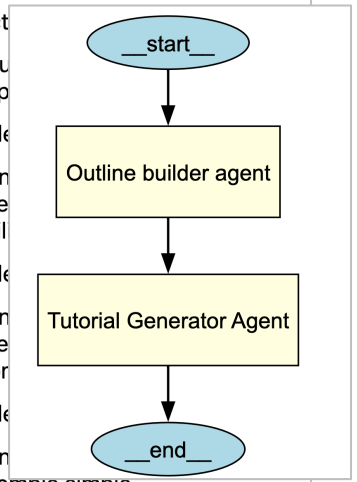
```python
display(Markdown("### Outline :"))
display(Markdown(tuto.outline))
print("*"*60)
display(Markdown("### Totorial :"))
display(Markdown(tutorial_response.final_output.tutorial))
```

```python
draw_graph(outline_builder_agent)
```

```python
outline_builder_agent = Agent(
    name= "Outline builder agent",
    instructions="""
        Given a particular programming topic, your joc is to generate a tutorial by crafting an ouline
        After making the outline, hand it to the tutorial generator agent"
    """,
    handoffs=[tutorial_generator_agent]
)
```

Outline :

1. Introduct
   - Pou
   - Typ
2. La boucle
   - Syn
   - Exe
   - Util
3. La boucle
   - Syn
   - Exe
   - Cor
4. La boucle
   - Syn
   - Exemple simple
   - Différences avec `while`



Triage Agent

```python
from agents import Agent, handoff, RunContextWrapper, function_tool
from agents.extensions.visualization import draw_graph
```

```python
@function_tool
def get_teacher_name(course : str):
    """
    Get The teacher name if course (Mathematics, History)
    """
    if(course=='Mathe
        return "Pr. H
    else:
        return "Pr. Najlaa"
```

(parameter) course: str

course (Mathematics, History)

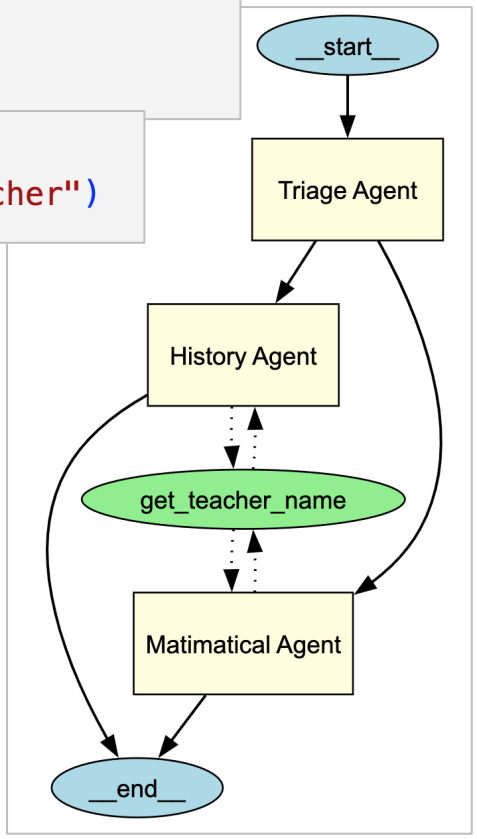
```python
triage_agent = Agent(
    name= "Triage Agent",
    instructions="""
        Answer the user question by delegating the task to the apropriate agent,
        If neither agent is relevant, give a general response
    """,
    handoffs=[history_agent, math_agent]
)
```

```python
history_agent = Agent(
    name= "History Agent",
    handoff_description= "Specialist Agent for historical questions",
    instructions="""
        You provide asistant with the historical queries
        Explain important events and context clearlly
    """,
    tools= [get_teacher_name]
)
```

```python
math_agent = Agent(
    name= "Matimatical Agent",
    handoff_description= "Specialist Agent for matimatical questions",
    instructions= """
        You provide asistant with the matimatical queries
        Explain your reasoning at each step and include examples
        """,
    tools= [get_teacher_name]
)
```

```python
trial_response = await Runner.run(triage_agent,
        "How do i multiply X by y and give me the name of teacher")
```

```python
draw_graph(triage_agent)
```



Triage Agent

```python
from agents import Agent, handoff, RunContextWrapper, function_tool
from agents.extensions.visualization import draw_graph
```

```python
@function_tool
def get_teacher_name(course : str):
    """
    Get The teacher name if course (Mathematics, History)
    """
    if(course=='Mathe
        return "Pr. H
    else:
        return "Pr. Najlaa"
```

(parameter) course: str

course (Mathematics, History)

```python
triage_agent = Agent(
    name= "Triage Agent",
    instructions="""
        Answer the user question by delegating the task to the apropriate agent,
        If neither agent is relevant, give a general response
    """,
    handoffs=[history_agent, math_agent]
)
```

```python
history_agent = Agent(
    name= "History Agent",
    handoff_description= "Specialist Agent for historical questions",
    instructions="""
        You provide asistant with the historical queries
        Explain important events and context clearlly
    """,
    tools= [get_teacher_name]
)
```

```python
math_agent = Agent(
    name= "Matimatical Agent",
    handoff_description= "Specialist Agent for matimatical questions",
    instructions= """
        You provide asistant with the matimatical queries
        Explain your reasoning at each step and include examples
        """,
    tools= [get_teacher_name]
)
```

```python
trial_response = await Runner.run(triage_agent,
        "How do i multiply X by y and give me the name of teacher")
```

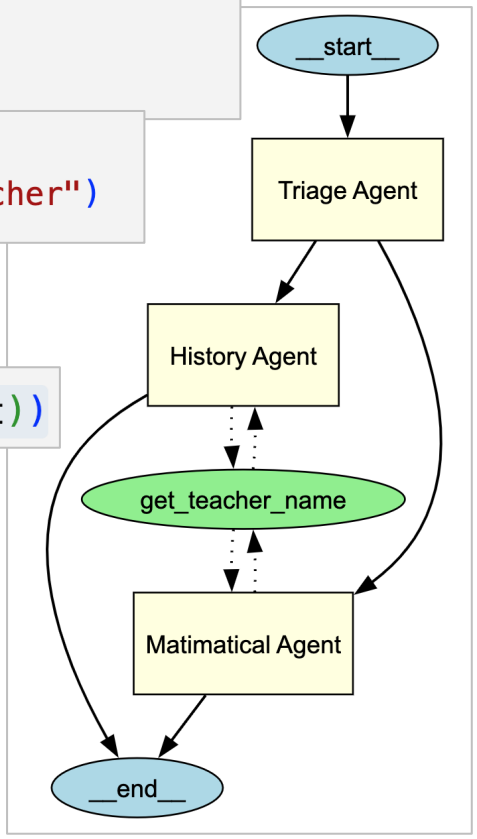
```python
draw_graph(triage_agent)
```

```python
display(Markdown(trial_response.final_output))
```

To multiply two numbers, (x) and (y), follow these steps:

1. **Understand the Multiplication:** Multiplication is a shortcut for repeated a
2. **Perform the Multiplication:**
   - If both (x) and (y) are positive integers, multiply them directly.
   - For example, if (x = 3) and (y = 4), then (3 \times 4 = 12).
3. **Use a Calculator for Complex Numbers:**
   - For large or decimal numbers, you might want to use a calculator.
4. **Special Properties:**
   - **Zero Property:** Any number multiplied by 0 is 0.
     - E.g., (5 \times 0 = 0).
   - **Identity Property:** Any number multiplied by 1 remains unchanged.
     - E.g., (7 \times 1 = 7).

The Mathematics teacher is Pr. Hassan.



Triage Agent

```
def on_history_handoff(ctx : RunContextWrapper[None]):
    print ("="*60)
    print ("History Agent is Invoqued")
    print ("="*60)

def on_math_handoff(ctx : RunContextWrapper[None]):
    print ("*"*60)
    print ("Math Agent is Invoqued")
    print ("*"*60)
```

```
triage_agent = Agent(
    name= "Triage Agent",
    instructions="""
        Answer the user question by delegating the task to the apropriate agent,
        If neither agent is relevant, give a general response
    """,
    handoffs=[
        handoff(history_agent, on_handoff=on_history_handoff),
        handoff(math_agent, on_handoff=on_math_handoff)
            ]
)
```

```
trial_response = await Runner.run(triage_agent,"HISTORY OF Rabat")
```

```
============================================================
History Agent is Invoqued
============================================================
```

```
trial_response = await Runner.run(triage_agent,"C'est quoi l'ENSET")
display(Markdown(trial_response.final_output))
```

L'ENSET (École Normale Supérieure de l'Enseignem
disciplines techniques et professionnelles. Elle est s
futurs enseignants à transmettre des connaissance
programmes peuvent inclure des études théorique

```
trial_response = await Runner.run(triage_agent,"How do i multiply X by y")
```

```
************************************************************
Math Agent is Invoqued
************************************************************
```

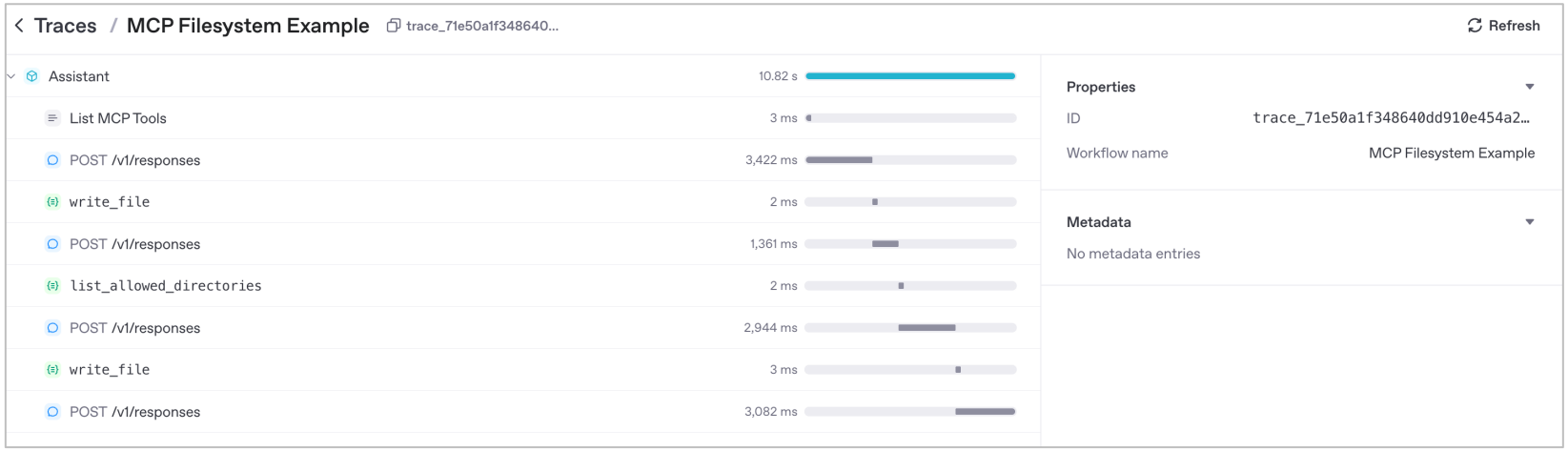
Agent Traces

https://platform.openai.com/traces

```
from agents import trace
```

```
with trace("Triage Work Flow"):
    trial_response = await Runner.run(triage_agent,"Hitoire de Rabat")
    trial_response.final_output
```

```
==========================================================
History Agent is Invoqued
==========================================================
```



Agent Streaming

```python
from openai.types.responses import ResponseTextDeltaEvent
joke_agent = Agent(
    name="My Agent",
    model='gpt-4o',
    instructions="You are a joke teller. you are giving a topic and you will tell 10 jokes about it"
)
result =  Runner.run_streamed(joke_agent,"Food")
async for event in result.stream_events():
    if event.type=='raw_response_event' and isinstance(event.data, ResponseTextDeltaEvent):
        print(event.data.delta, end="", flush=True)
```

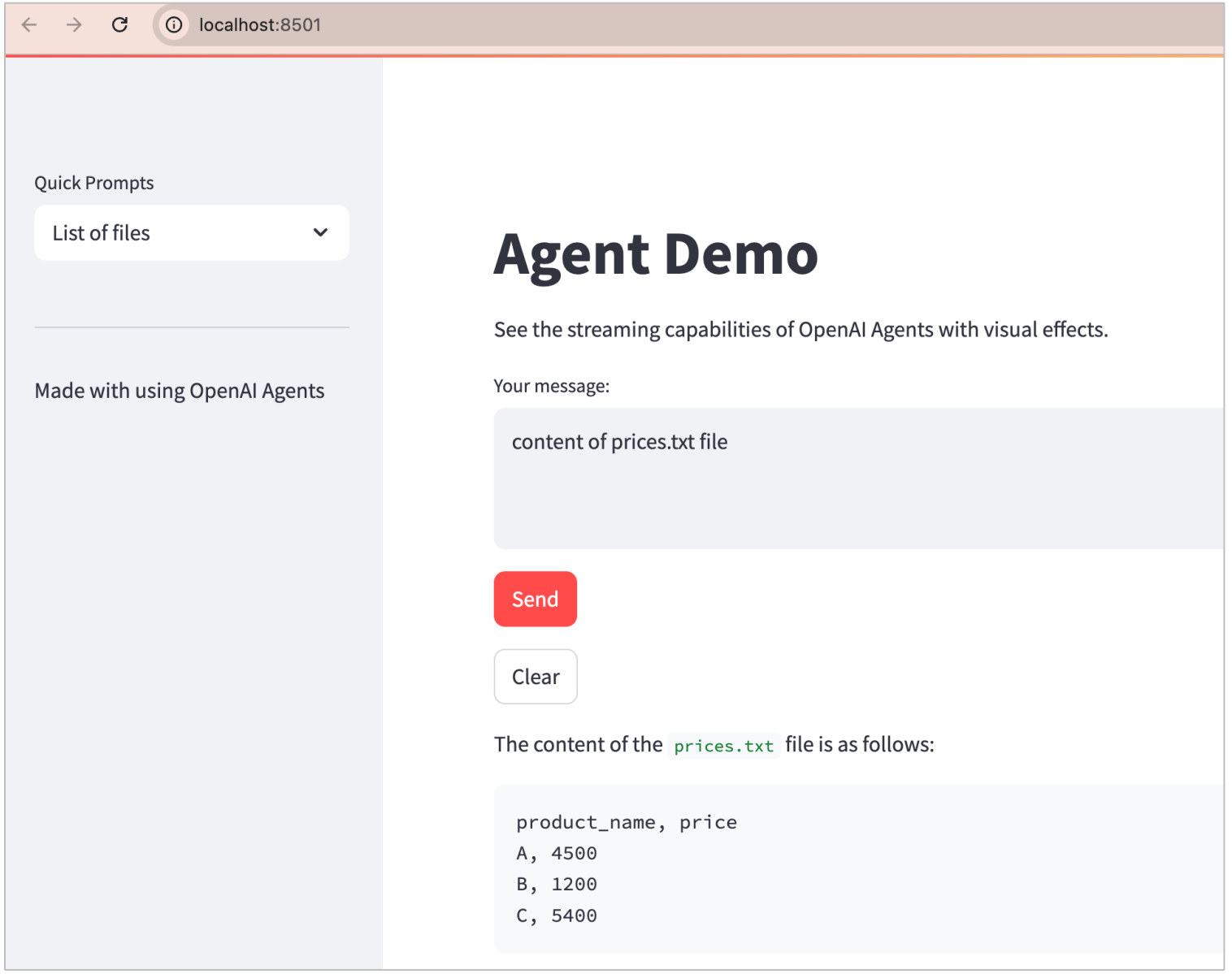
```
Sure thing! Here are 10 food jokes for you:

1. Why did the tomato turn red?
   Because it saw the salad dressing!

2. What did the taco say to the burrito?
   "I'm nacho friend anymore!"

3. Why don't eggs tell jokes?
   They might crack up!

4. What do you call cheese that's not yours?
   Nacho cheese!
```

Agent MCP : Model Context Protocol

```
## Cretae agent_mcp.py File
## Then Exceute this application in console : > python
agent_mcp.py
import asyncio
import os
import shutil
from agents import Agent, Runner, gen_trace_id, trace
from agents.mcp import MCPServer, MCPServerStdio
from dotenv import load_dotenv

load_dotenv()

async def run(mcp_server: MCPServer):
    agent = Agent(
        name="Assistant",
        instructions="Use the tools to read the filesystem and
answer questions based on those files.",
        mcp_servers=[mcp_server],
    )

    # List the files it can read
    message = "Read the files and list them."
    print(f"Running: {message}")
    result = await Runner.run(starting_agent=agent, input=message)
    print(result.final_output)
```

```
# Ask about books
    message = "What is my #1 favorite book?"
    print(f"\n\nRunning: {message}")
    result = await Runner.run(starting_agent=agent, input=message)
    print(result.final_output)

    # Ask a question that reads then reasons.
    message = "Look at my favorite songs. Suggest one new song that I might
like."
    print(f"\n\nRunning: {message}")
    result = await Runner.run(starting_agent=agent, input=message)
    print(result.final_output)

async def main():
    current_dir = os.path.dirname(os.path.abspath(__file__))
    samples_dir = os.path.join(current_dir, "sample_files")
    os.makedirs(samples_dir, exist_ok=True)
    async with MCPServerStdio(
        name="Filesystem Server, via npx",
        params={
            "command": "npx",
            "args": ["-y", "@modelcontextprotocol/server-filesystem",
samples_dir],
        },
    ) as server:
        trace_id = gen_trace_id()
        with trace(workflow_name="MCP Filesystem Example", trace_id=trace_id):
            print(
                f"View trace:
https://platform.openai.com/traces/trace?trace_id={trace_id}\n"
            )
            await run(server)

if __name__ == "__main__":
    # Let's make sure the user has npx installed
    if not shutil.which("npx"):
        raise RuntimeError(
            "npx is not installed. Please install it with `npm install -g npx`."
        )

    asyncio.run(main())
```

Agent MCP : Model Context Protocol with Streamlit UI



Agent MCP : Model Context Protocol with Streamlit UI

```
import asyncio
import streamlit as st
from openai.types.responses import ResponseTextDeltaEvent
from agents.mcp import MCPServer, MCPServerStdio, MCPServerSse
import sys
import os
from dotenv import load_dotenv
load_dotenv()
api_key = os.environ.get("OPENAI_API_KEY")
if not api_key:
    raise ("API KEY not available")
# Add the parent directory to the path so we can import the agents
module
sys.path.append(os.path.dirname(os.path.dirname(os.path.abspath(__fil
e__))))
from agents import Agent, Runner
st.set_page_config(
    page_title="Agent Demo",
    page_icon=".",
    layout="wide",
)
st.title("Agent Demo")
st.write("See the streaming capabilities of OpenAI Agents with visual
effects.")
```

```
with st.sidebar:
    demo_options = [
        "List of files",
        "if not created, create new file data.txt",
        "content of data.txt file",
        "Look at my favorite songs. Suggest one new song that I might like.",
    ]
    demo_prompt = st.selectbox("Quick Prompts", demo_options)
    st.markdown("---")
    st.markdown("Made with  using OpenAI Agents")
# User input area
user_input = st.text_area("Your message:", value=demo_prompt, height=100)
send_button = st.button("Send", type="primary")
# Response area with custom styling
response_container = st.container()
message_placeholder = st.empty()
```

Agent MCP : Model Context Protocol with Streamlit UI

```
import asyncio
import os
import shutil
from agents import Agent, Runner, gen_trace_id, trace
from agents.mcp import MCPServer, MCPServerStdio
from dotenv import load_dotenv
load_dotenv()
async def run(mcp_server: MCPServer):
    agent = Agent(
        name="Assistant",
   instructions="answer the user question using provided tools",
        mcp_servers=[mcp_server],
    )
    if send_button and user_input:
        with response_container:
            with st.spinner("Thinking..."):
                result = await Runner.run(starting_agent=agent,
input=user_input)
                print(result.final_output)
                message_placeholder.markdown(result.final_output)
            # Add clear button
            if st.button("Clear"):
                st.experimental_rerun()
```

```
# Instructions
    if not send_button:
        with response_container:
        st.info("👆 Enter your message above and click 'Send' to see the streaming response.")
            st.markdown(
                """
            ### Tips:
            - Choose from the quick prompts or enter your own
            """)
async def main():
    current_dir = os.path.dirname(os.path.abspath(__file__))
    samples_dir = os.path.join(current_dir, "sample_files")
    os.makedirs(samples_dir, exist_ok=True)
    async with MCPServerStdio(
        name="Filesystem Server, via npx",
        params={
            "command": "npx",
            "args": ["-y", "@modelcontextprotocol/server-filesystem", samples_dir],
        },
    ) as server:
        trace_id = gen_trace_id()
        with trace(workflow_name="MCP Filesystem Example", trace_id=trace_id):
            print(
                f"View trace: https://platform.openai.com/traces/trace?trace_id={trace_id}\n"
            )
            await run(server)
if __name__ == "__main__":
    # Let's make sure the user has npx installed
    if not shutil.which("npx"):
        raise RuntimeError(
            "npx is not installed. "
        )
    asyncio.run(main())
```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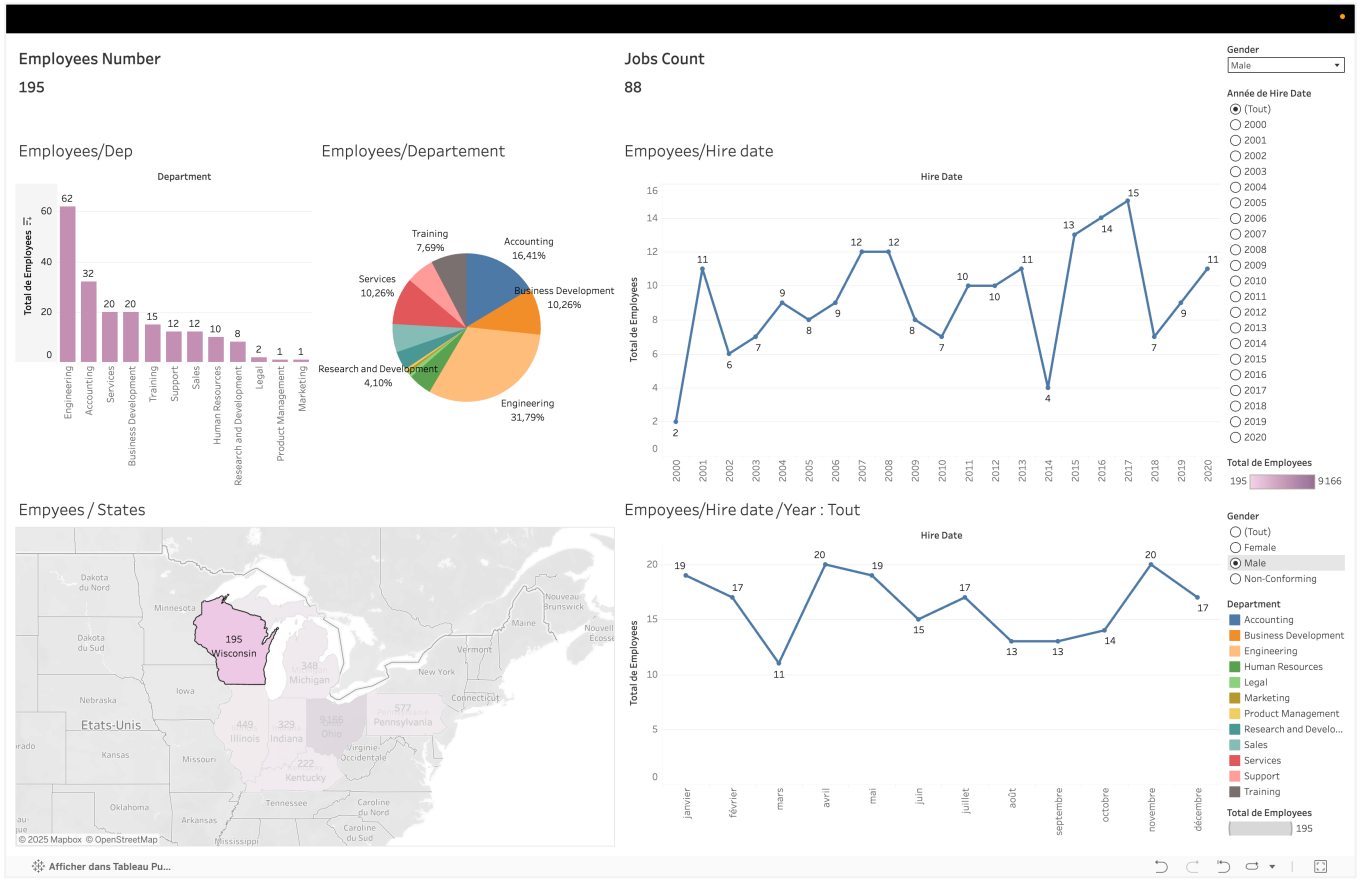
Data Analytics With Tableau Public

Mohamed Youssfi, Enseignant Chercheur, ENSET Mohammedia, Directeur Laboratoire Informatique, Intelligence Artificielle et Cyber Sécurité



+ableau from Salesforce

+ableau public